SAINT MICHAEL COLLEGE OF CARAGA

EVENT EVALUATION AND ATTENDANCE SYSTEM

FOR SAINT MICHAEL COLLEGE OF CARAGA

A Capstone Project Presented To

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SAINT MICHAEL COLLEGE OF CARAGA

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CHAPTER 1

INTRODUCTION

1.1 Project Context

Events are significant gatherings that provide opportunities for people to come together to learn, collaborate, engage, and share ideas. Participating in certain Events also offers activities that contribute to an individual's personal growth and development. However, the process of recording attendance and collecting evaluations in such events is often done manually using paper and pen. This manual approach is time-consuming, prone to errors, and difficult to manage when handling large groups from different departments or year levels. As a result, organizers face challenges in monitoring attendance, preparing reports, and ensuring that evaluations are complete.

According to Abd El-Mawla et al. [1], many schools and organizations still use paper to track attendance, which requires a significant amount of time and often leads to errors when counting attendance sheets, making it hard to manage attendance, particularly when dealing with large groups of people. Furthermore, they only dealt with attendance taking and did not address event evaluation or the preparation of detailed reports.

In addition, M. A. D. Perin et al. [2] highlight that traditional methods of monitoring attendance, such as manual sign-ins and logbooks, are often time-consuming and prone to human error. Keeping track of accurate records becomes harder for schools and organizations when handling a larger number of attendees. Moreover, their work focused on the efficiency of recording attendance but did not include the evaluation of events or the generation of detailed reports.

Traditionally, attendance and event evaluation have been done manually on paper forms in schools and organizations. At Saint Michael College of Caraga, this practice has been common

during various activities and events. While it serves its purpose, it often leads to slow and tedious processes, as students must wait in line to sign in. Manual entries can result in inaccurate information, as there is no centralized record-keeping system to track attendance easily. Consequently, those who consolidate the results may also make mistakes.

To address the challenges of manual attendance and evaluation at SMCC, a web-based system utilizing QR codes is proposed. The event will be created by administrators, department heads, and personnel, and students can scan their school ID with a QR code and record their attendance with a little effort, as QR codes are widely known as a fast and efficient way to track attendance, improve data accuracy, and provide a positive user experience[3]. The system will enable the online submission of evaluations and automatically generate reports and certificates. This approach will streamline attendance and assessment, making them faster, easier, and more accurate, while reducing errors and physical contact.

The Event Evaluation and Attendance System will utilize a QR code that helps administrators and department heads to manage events more conveniently by generating events, monitoring attendance, and automatically producing reports. Staff can keep an eye on participation with accuracy, and students can complete evaluations and obtain certificates without delay. This will help automate manual work, reduce errors, save time, and provide credible records, improving overall efficiency and supporting better management of school events.

1.2 Literature Review

This chapter presents the relevant literature that backs up the researchers' investigation into the current issues. It highlights the necessity of implementing the suggested software application, drawing attention to both its benefits and the challenges it aims to address. The reviewed literature is trustworthy because it originates from reputable online journals, peer-reviewed articles, e-books, electronic references, and other web-based resources.

QR Code-Based Attendance Systems

Khan and Singh [4], David et al. [5], Deshmukh et al. [6], Nuralif and Fachrie [7], Ibrahim and Talfan [8], Mar'atuttahirah et al. [9], Choudhary et al. [10], Garrido [11], Jonathan [12], Elewaily et al. [13], Mukhtar et al. [14], Santos et al. [15], and Wenny et al. [16] highlighted the effectivess and usefullness. Through features such as dynamic code generation, encryption, multi-factor authentication, WhatsApp integration, simulation-based redesign, and optimization algorithms, their work consistently demonstrates how QR codes enhance security, streamline attendance, and reduce manual errors. The system is being utilized across diverse contexts, including universities in Uganda, Sudan, and Pakistan, as well as schools in the Philippines and high school classrooms. They consistently demonstrated scalability, usability, and strong user acceptance. Collectively, these writers indicate that QR code-based systems are the most beneficial technology for educational and professional settings, as they provide efficient, safe, and flexible methods to automate attendance.

In contrast, Gomes et al. [17] and Aware et al. [18] explored facial recognition technologies for attendance management. Gomes employed Haar Cascade and LBPH algorithms to detect multiple faces simultaneously, while Aware used deep learning to authenticate identities. Despite their intention to eradicate fraudulent attendance, these systems created privacy issues and necessitated the development of complex infrastructure and the purchase of expensive equipment. Facial recognition techniques are more costly and require more technical know-how than QR systems, which restricts their scalability in institutions with limited resources. These authors demonstrate that while facial recognition offers advanced identity verification, its complexity and cost make QR code-based systems the more practical and scalable choice for institutions that prioritize simplicity, speed, and efficiency.

Smart Attendance Systems

Innovative attendance systems were created by Hamzah et al. [19] and Albahrani et al. [20] to overcome the shortcomings of manual tracking. Hamzah's system utilized QR codes to automate attendance and generate reports, emphasizing ease of use and accessibility for educational and event contexts. To inspire confidence in selecting suitable solutions, Albahrani's Smart Attendance Management System (SAMS) integrates face recognition, fingerprint scanning, and GPS location, prioritizing multi-factor authentication and location-based verification. Together with RFID-based methods, such as those presented by Yadav et al. [21] and the systematic review by Ishaq & Bibi [22], these studies demonstrate how innovative attendance systems can significantly enhance data accuracy, reduce administrative burden, and enable real-time monitoring—capabilities that align with the objectives of contemporary event management platforms.

However, the systems differ in their technical complexity. Hamzah emphasized user convenience and accessibility, which can make operational tasks feel more straightforward and more intuitive. In contrast, Albahrani focused on advanced security protocols, offering reassurance about data safety. Similarly, RFID-based systems highlight their cost-effectiveness and portability, helping users feel confident about affordability and flexibility. Meanwhile, systematic reviews highlight challenges in scalability and data security, prompting users to consider potential concerns. This contrast suggests that innovative attendance systems can be tailored to specific operational needs, whether prioritizing usability, affordability, or robust identity verification. Systems designed for event evaluation may benefit from adopting a balanced approach that ensures both usability and data integrity.

QR Code and Face Recognition-Based Attendance Systems

Winanti et al. [23] and Agripa & Astillero [24] implemented QR code-based systems to automate attendance and reporting, showcasing features like contactless check-ins and offline QR code generation. Riza and Warif [25] introduced a more secure QR code system utilizing the CAOQTP algorithm, which combines OTP and CAPTCHA to prevent fraud and restricts QR scanning to specific physical locations, demonstrating the flexibility and safety of QR codes for attendance tracking. These features make QR code systems suitable for various applications requiring fraud prevention and operational flexibility.

Nurkhamid et al. [26] developed a facial recognition-based attendance system using deep learning, achieving 81.25% accuracy when students faced forward; however, this accuracy declined under poor lighting or varied positioning. Sunaryono et al. [27] proposed an Android-based attendance system that combines QR code verification with face recognition, achieving 97.29% accuracy using linear discriminant analysis and minimizing processing time to 0.000096 seconds. Dang [28] further advanced this concept with an improved FaceNet model, based on MobileNetV2 and SSD, which achieves 95–99% accuracy and 25 FPS, optimized for low-resource mobile and IoT devices. Sriyanto et al. [29] designed a machine learning-based facial recognition system for distance learning, integrating LBPH and KNN algorithms, which achieved up to 100% accuracy under ideal conditions but showed sensitivity to lighting and camera quality. Essien & Ansa [30] proposed a deep learning-based attendance system using MTCNN for detection and Deepface with FaceNet-512 for recognition, demonstrating high accuracy but requiring large datasets and substantial computational resources. These studies highlight that combining methods can enhance performance and reliability, inspiring confidence in hybrid biometric solutions that adapt to environmental challenges and hardware limitations.

1.3 Objective of the Study

The researchers' study aims to design, develop, evaluate, and implement an Event Attendance and Evaluation System for Saint Michael College of Caraga.

Specifically, it aims to:

1. Develop an event management module that allows authorized users to efficiently create, manage, and track events.
2. Provide a dashboard for viewing and managing attendance and evaluation records with restricted access based on user roles and departments.
3. Implement QR code technology to easily record attendance of students and personnel during events.
4. Generate evaluation results, reports, and certificates automatically for each event to ensure accuracy and save time.
5. Evaluate the system's functionality, efficiency, and usability using ISO 25010 standards.

1.4 Scope and Limitations

Saint Michael College of Caraga is the exclusive user of the Event Evaluation and Attendance System. The system has four user roles: department heads, who can create events, plot event, evaluate students, view attendance and evaluation records and can print and view reports only for their departments; personnel, who can view student attendance records within their department; personnel and students, can log in to evaluate events they attended, receive a certificate of attendance, and use their mobile devices to scan their school ID QR code for attendance; and the superuser (admin), who has full access to create and manage all events, view all attendance and evaluation records, manage account and generate reports.

Since the system only functions within the school's secure network, it cannot be accessed by the general public or from outside the school grounds. To complete assessments and record attendance, users must be physically present at the school. Department heads cannot view records outside of their own department because access to data is limited based on user roles. The school ID QR code is the only code accepted for tracking attendance, as it utilizes QR codes embedded in the school IDs. Only mobile devices with cameras can use the QR code scanning feature, ensuring an efficient and automated attendance tracking process.

1.5 Definition of Terms

The terms defined below explain the basic concepts and factors discussed during the conceptualization and development of the Event Evaluation and Attendance System, and contextual information on how the researchers apply these concepts in their study.

Back-end – In this system, the back-end serves as the system core that handles data processing and storage. It is responsible for tracking attendance logs, saving event details, processing evaluation results, and generating digital certificates. This component ensures that all user actions performed on the front-end are securely and accurately recorded in the system database.

Dashboard – The dashboard is a responsive interface within the system that provides users with access to real-time summaries, attendance, and evaluation reports. It supports decision-making by offering an integrated view of event participation and feedback results.

Front-end – Refers to the visual and interactive elements that users engage with, including forms, buttons, menus, and layout designs. Built using HTML, CSS, and JavaScript, this layer allows users, such as students and personnel, to log attendance, submit evaluations, and view event-related information in a user-friendly format.

QR Code (Quick Response Code) – In this study, a QR code refers to a two-dimensional bar code that stores information such as student and personnel ID numbers. The QR code will be used by students and personnel to scan their attendance.

QR Code Scanner –In this system, the QR code Scanner refers to a device or mobile camera that scans the QR code of the school ID. Within the system, this tool plays a crucial role, that automates the recording of the attendance of all the students and personnel.

Records Management – This feature enables authorized users to store, organize, and retrieve attendance and evaluation data. It supports search functionality and report generation, making it easier for stakeholders to monitor participation trends and assess event effectiveness.

Stakeholders – refers to the users involved in the operation of the system, including students, school personnel, and department heads. Stakeholders will benefit from the fast process of attendance and evaluation.

Super Admin – This refers to the highest user role in the system, where this user can manage accounts, oversee the system, and access all attendance and evaluation records.

User Roles – The different types of access given to specific users that will be using the system: Super Admin, Department Head, Personnel, and Student. Each has different roles depending on their responsibilities.

Web-Based System – The system is designed as a web-based application accessible through browsers within the Saint Michael College of Caraga network. It requires internet connectivity and login credentials, ensuring that only authorized users can manage attendance, evaluations, and event records secure.

CHAPTER II

METHODOLOGY

2.1 Research Design

This study will use a Design and Development Research (DDR) approach, which focuses on creating and evaluating the Event Evaluation and Attendance System for Saint Michael College of Caraga as a technology-based solution to improve event evaluation performance.

2.2 Software Development Life Cycle



Figure 1. Agile Methodology

The Event Evaluation and Attendance System for Saint Michael College of Caraga will be developed by following the Agile Software Development Life Cycle and its principles of iterative development and stakeholder interaction. Agile methodologies prioritize flexibility by breaking the workflow into sprints or iterations. Rather than following a linear development and testing sequence, small functional portions of the system are built and tested in rapid succession. Event organizers and participants help gather requirements to identify critical features for the system during the initial planning phase. These features include user logging, attendance tracking, evaluation forms, and reports. Activities related to design, development, and testing will be

included in each module. Feedback from stakeholders is routinely gathered to aid in system improvement. Stakeholders will verify system design and development progress during the process to make sure user requirements are satisfied. Following development and deployment, agile approaches will assess ISO 25010 standards and offer continuous improvements based on user feedback. This strategy aids in making sure the system is user-centered and adaptable enough to meet the changing needs of the organization.

2.3 System Architecture

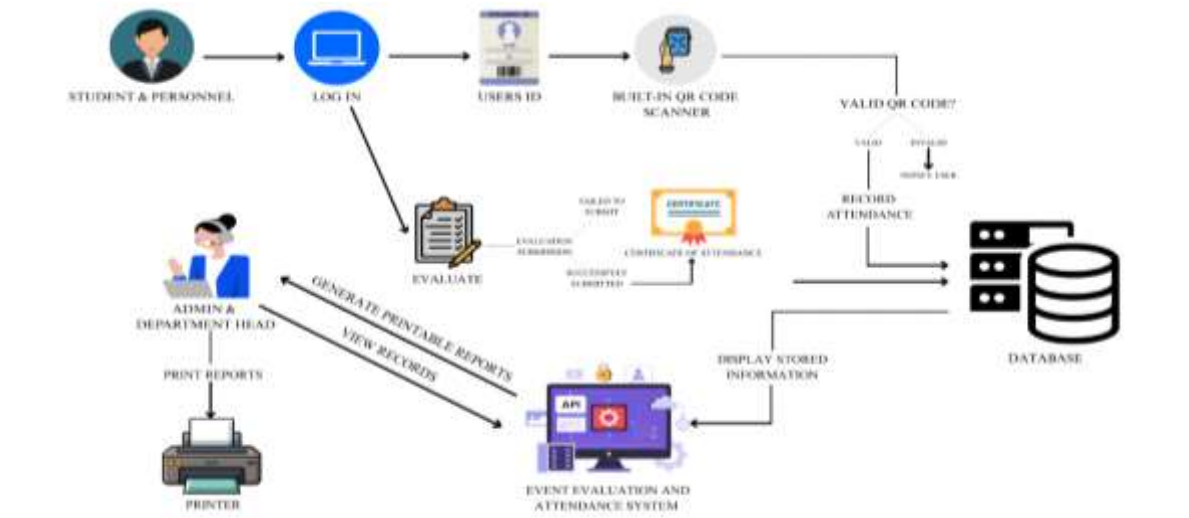


Figure 2. EEASFSMCC (System Architecture)

Figure 2 shows how to use the system in its entirety. After logging in, users must scan their school ID using the integrated QR code scanner, and the system logs their attendance by entering the time, date, and identification details into the database. Users then move on to the evaluation phase, where their input is gathered and arranged. The system then displays the stored data and automatically generates an attendance certificate. The administrator and department head can view all records, manage data, and generate printable reports via the system's interface. Only the administrator or superuser has complete access to the system.

2.4 Conceptual Framework

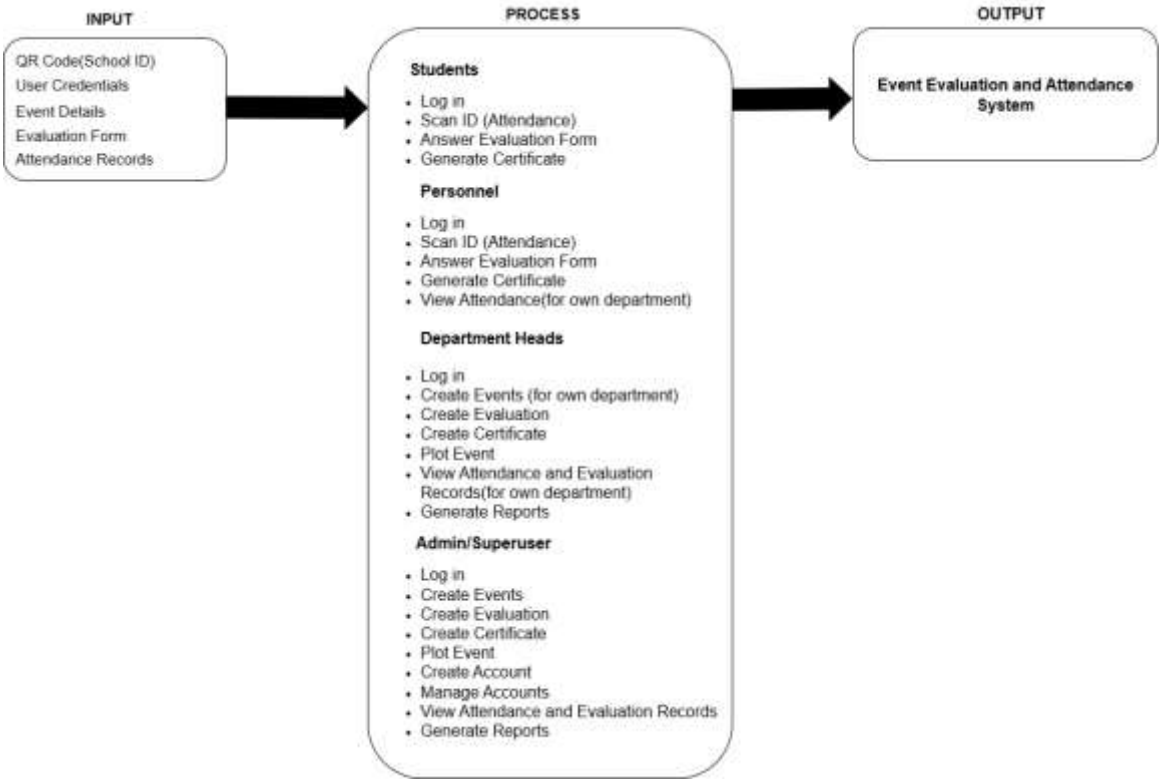


Figure 3. Input-Process-Output Diagram

Figure 3 displays the Input-Process-Output (IPO) Diagram for the proposed Event Evaluation and Attendance System. The procedure outlines the various actions that students, department heads, staff, and administrators take, and the input includes the user's credentials, evaluation form, event details, and QR code or school ID. Students can fill out the assessment form, create their certificate, log in, and scan their ID to verify their attendance. Employees can view department-specific attendance, log in, and scan their ID for attendance. Department heads have the ability to view attendance and reports, create events for their department, and log in. Super users or administrators have the ability to manage user accounts, log in, create events, keep an eye on attendance records, and produce reports.

2.5 Use Case Diagram

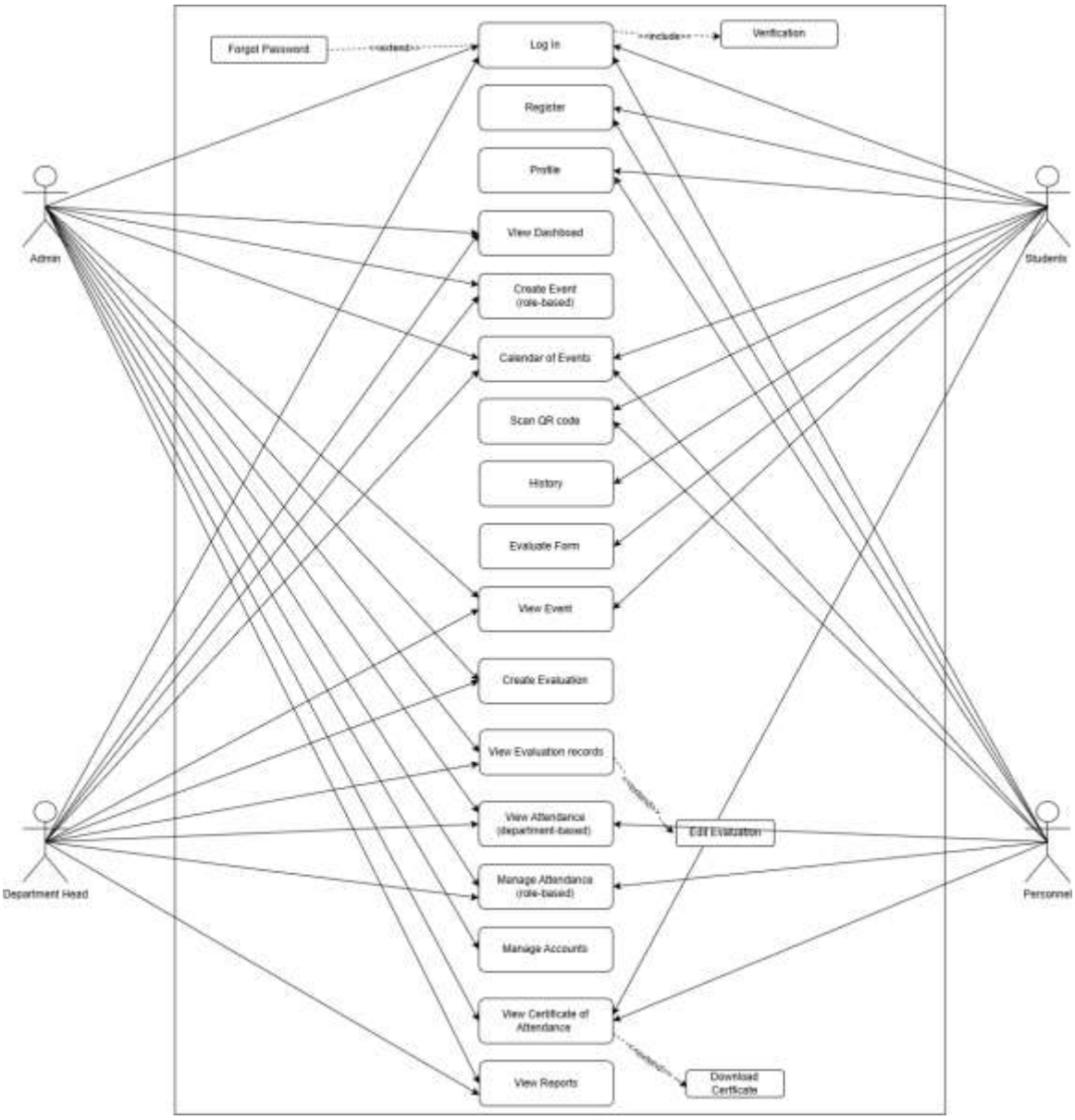


Figure 4. Use Case Diagram

Figure 4 shows the interaction between different types of users and the system. Students will verify if their school ID is valid. If valid, they will proceed to log in, scan their IDs for attendance, answer evaluation forms, and generate certificates. Forgot Password is optional if students happen to forget their password. Personnel can log in to the system to scan their IDs for attendance and view attendance specific to their department. Department heads are allowed to log in, create events, and access consolidated reports and attendance records specific to their

department. Meanwhile, the admin manages the system, including creating events, managing user accounts, tracking attendance records, and generating statistical reports.

2.6 Activity Diagram

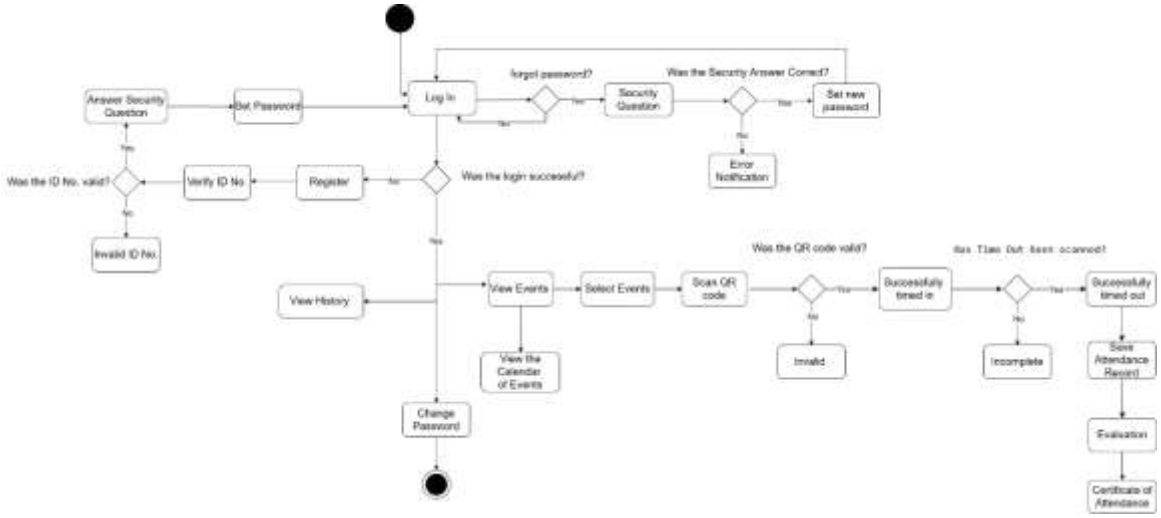


Figure 5. Activity Diagram-Student

Figure 5 illustrates the flow of activities that students perform when interacting with the system. Students are required to verify and set their passwords first before logging in. If a password is set, the student logs in with their credentials. If the login is successful, they will proceed; otherwise, they will be sent back to the login page. After that, they scan the QR code in the ID. To record their attendance, the system will validate the scanned QR code. If students forget the password, they must answer the security questions before they can reset the password to log in. Then, students can answer the evaluation form associated with the event they attended. Once the evaluation form is submitted, the system processes the data and allows the student to generate and download a certificate of attendance. Finally, the user can log out to end the session.

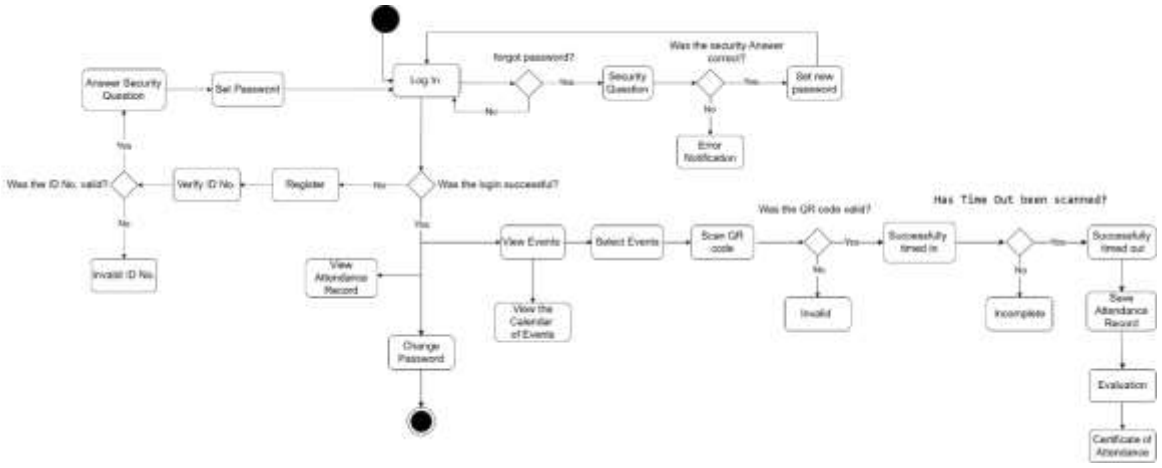


Figure 6. Activity Diagram- Personnel

Figure 6 illustrates the sequence of activities that personnel perform when interacting with the system. Personnel are required to verify and set their passwords first before logging in. If the login is successful, they will proceed; otherwise, they will be sent back to the login page. After that, they scan the QR code in the ID. To record their attendance, the system will validate the scanned QR code. If personnel forget their password, they must answer the security questions before they can reset their password. Then, personnel can view student attendance records specific to their departments. Finally, the user can log out to end the session.

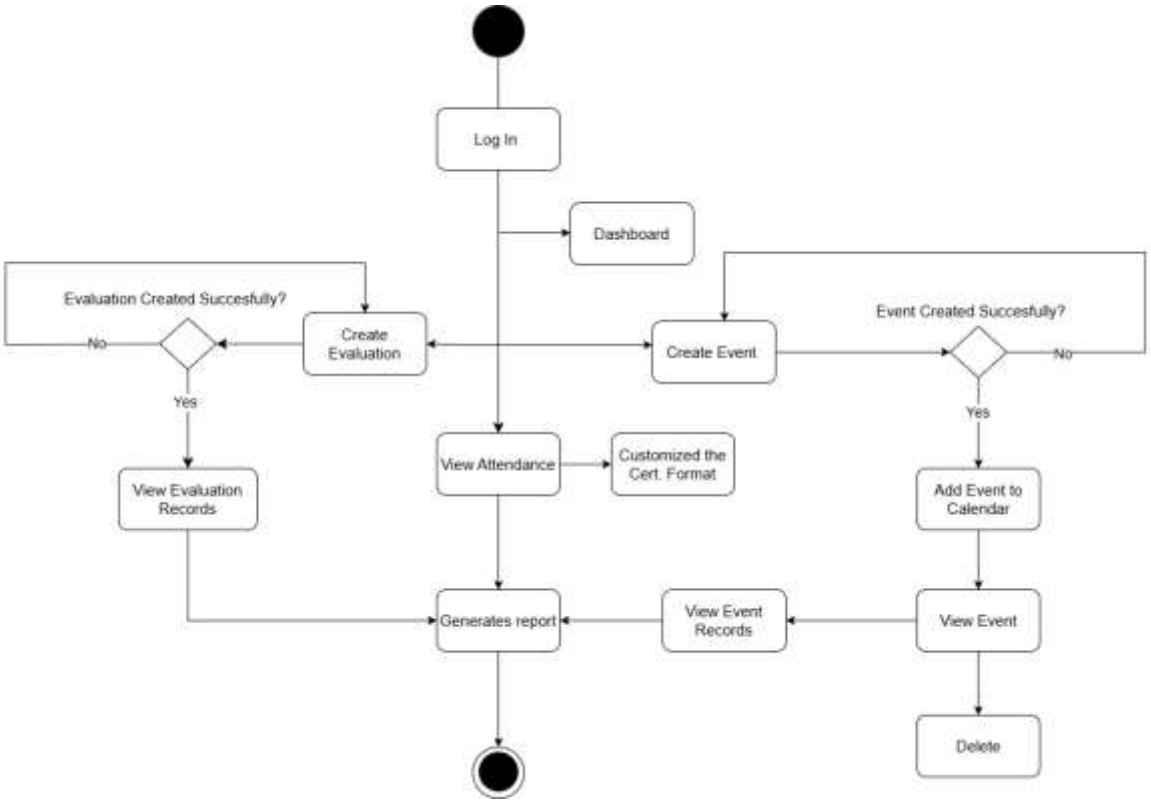


Figure 7. Activity Diagram -Department Head

Figure 7 illustrates the actions performed by department heads when using the system. The process begins with login. Once logged in, department heads can select various actions such as creating evaluations, viewing attendance, generating reports, or creating events. If an event is created successfully, they may proceed to view event records, manage existing events, or delete events(optional). Finally, the user can log out to end the session.

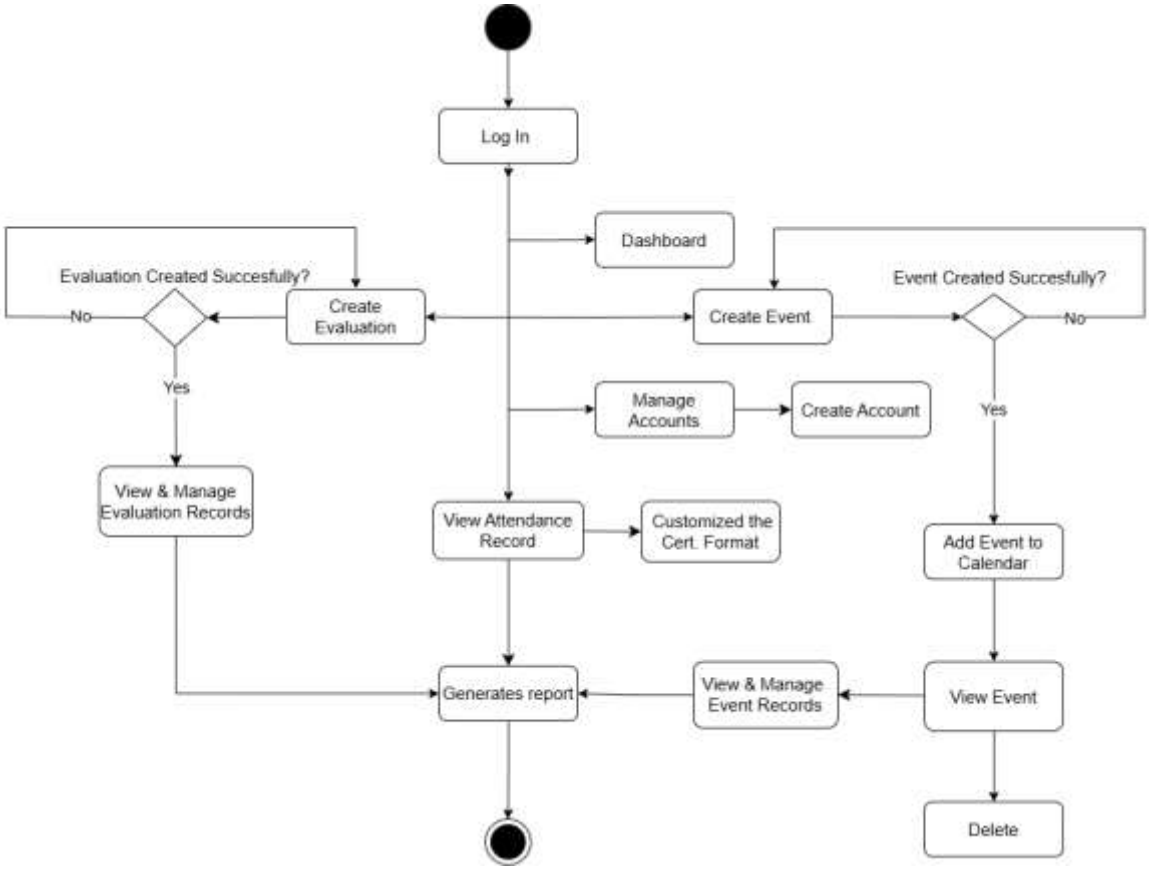


Figure 8. Activity Diagram- Admin

Figure 8 shows the workflow of the administrator, who has the highest level of system access. The process starts with login. Once logged in, the admin can select various actions such as creating evaluations, managing evaluation records, viewing and managing attendance records, and generating reports. Admin can create events, view and manage event records, view event details, or delete events when necessary. Finally, the user can log out to end the session.

Sequence Diagram

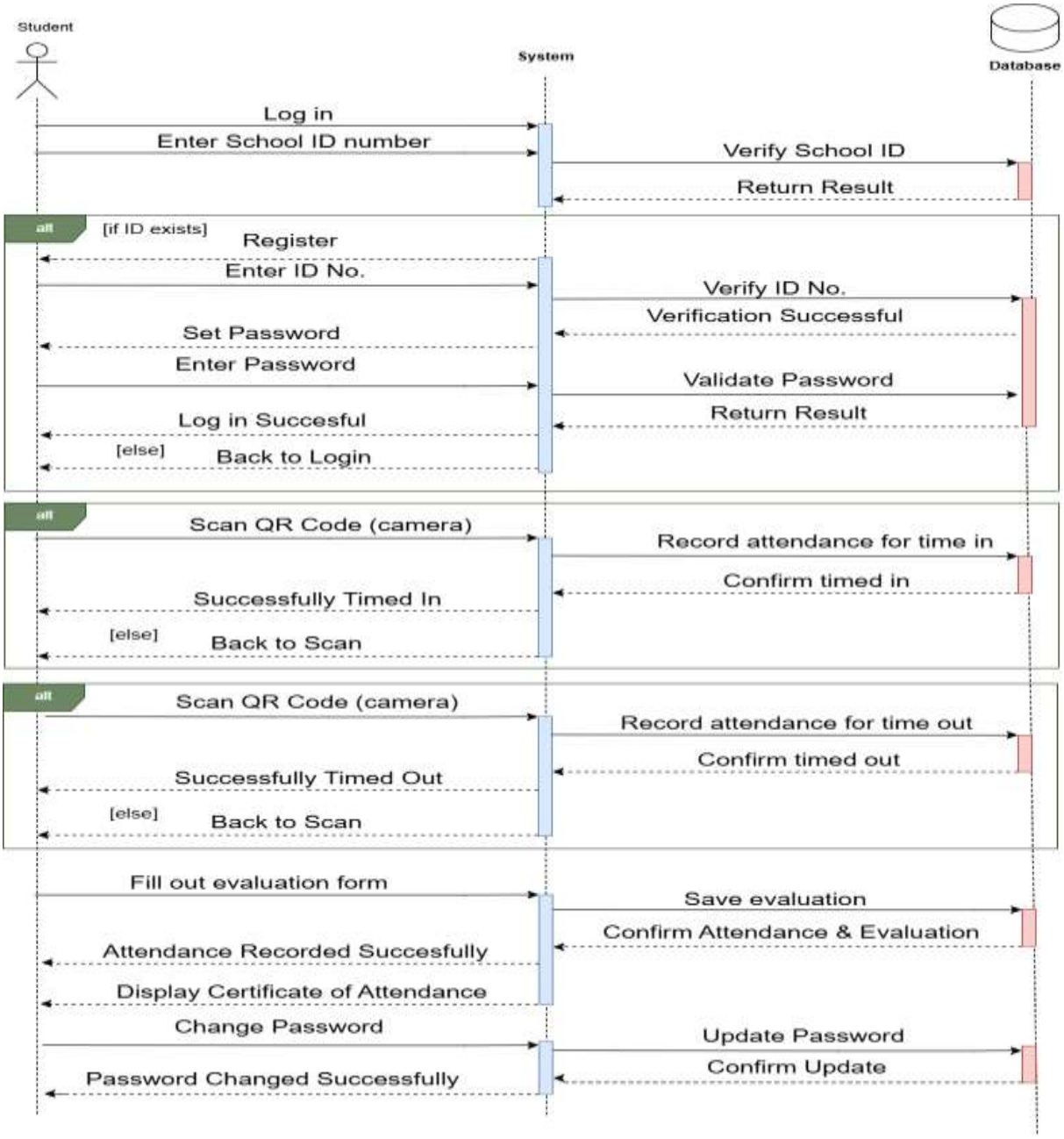


Figure 9. Sequence Diagram-Student

Figure 9 illustrates the interaction between the student, the system, and the database. Upon login, the system verifies the student's school ID. If valid, the student sets a password and gains access; otherwise, they are redirected to the login page. Once logged in, the student scans a QR code to record attendance, completes an evaluation form, and submits it. The system saves

both attendance and evaluation data to the database, then generates and displays the certificate of attendance.

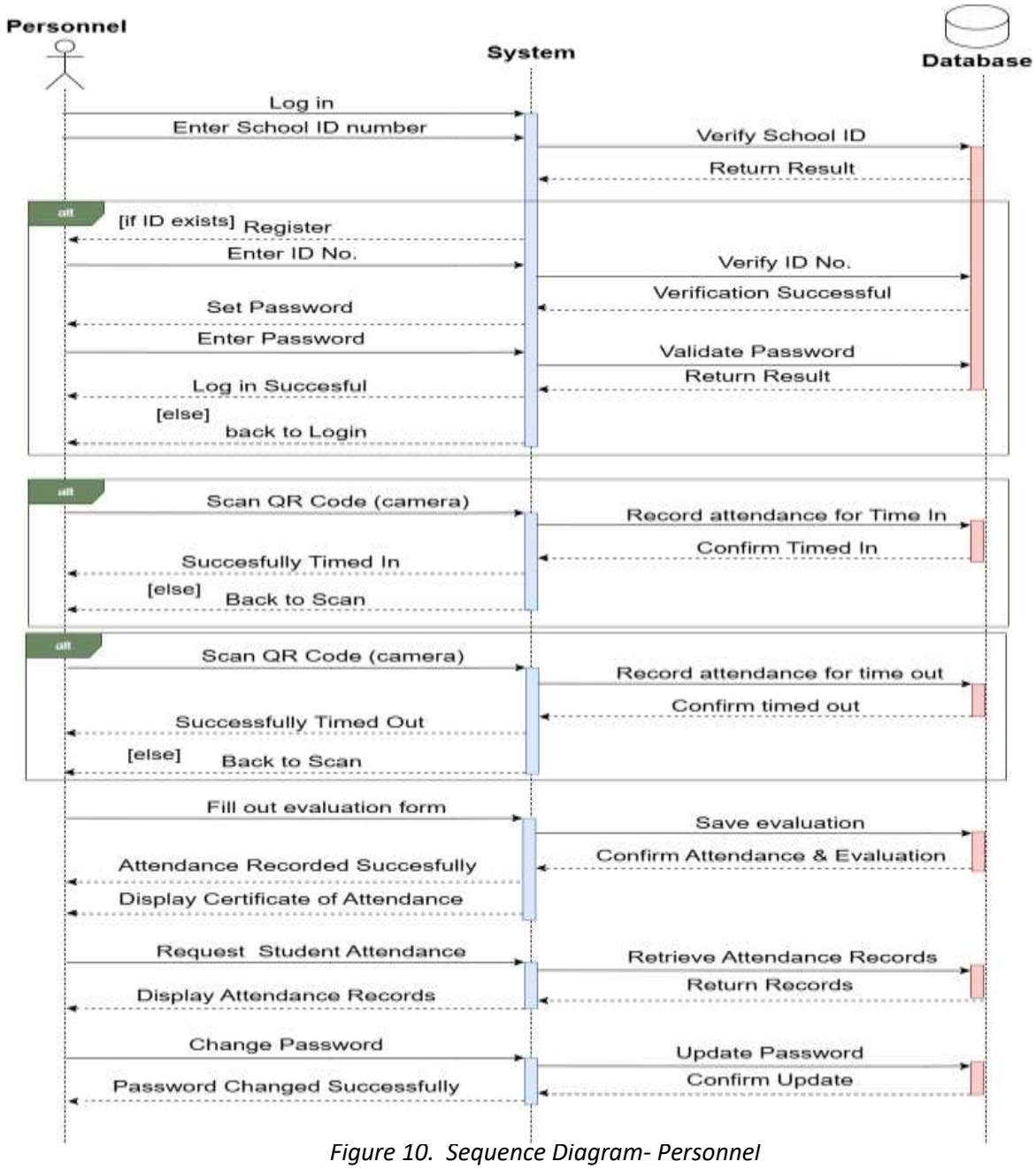


Figure 10. Sequence Diagram- Personnel

Figure 10 illustrates the interaction between the personnel, system, and database. The process begins with login authentication using a school ID, followed by password setup and verification. Once logged in, the personnel records attendance by scanning a QR code for time-in

and time-out, with each action confirmed by the system. They then complete an evaluation form, and the system saves both attendance and evaluation data. Finally, the system generates and displays the certificate of attendance.

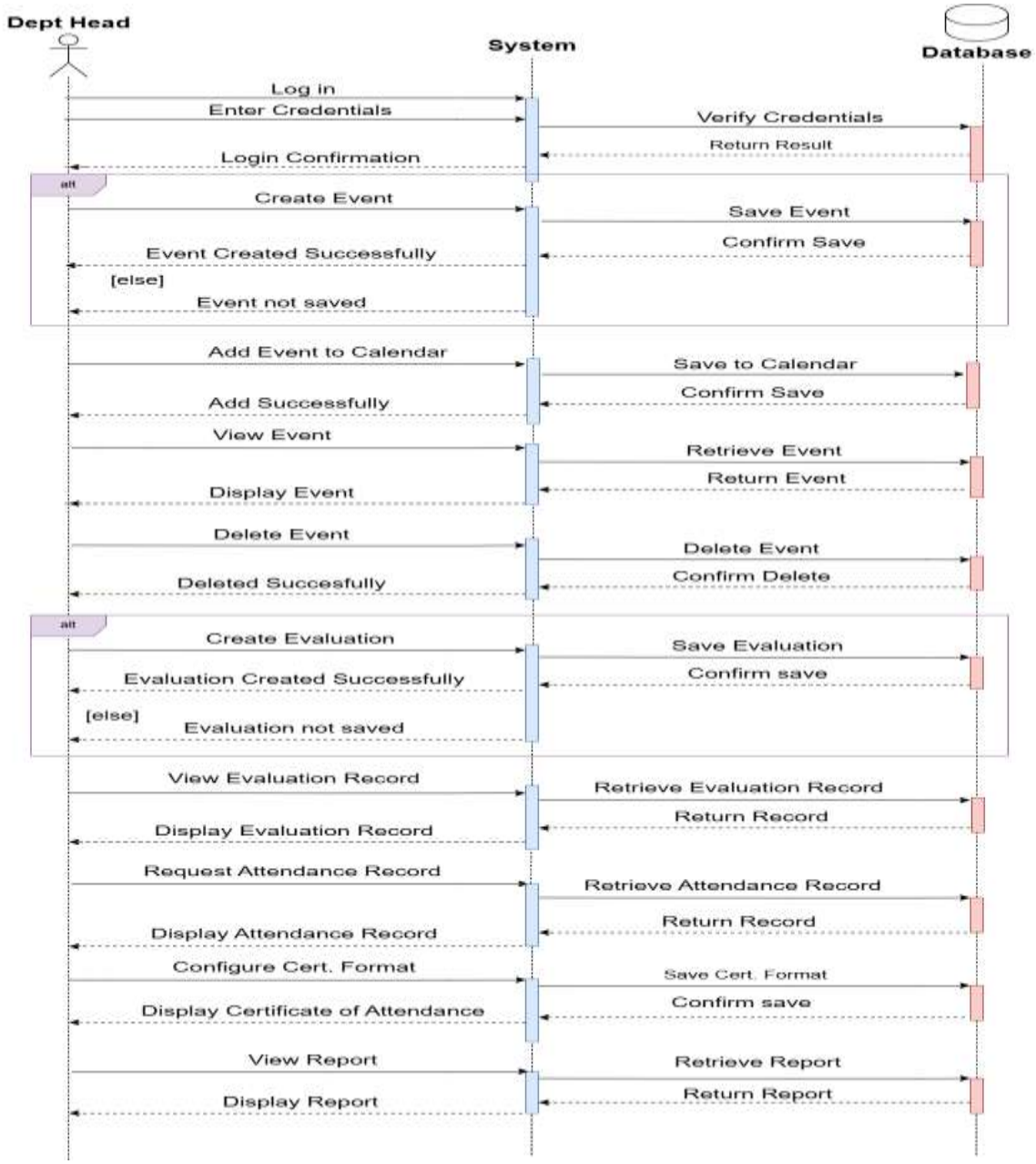


Figure 11. Sequence Diagram- Department Head

Figure 11 illustrates the interaction flow between the department head, the system, and the database. After logging in via credential verification, the department head can create, view,

and delete events, with each action processed and confirmed by the system. They can also generate evaluations, configure certificate formats, and access attendance, evaluation, and report records. All data transactions are handled through system-database exchanges to ensure accurate retrieval and storage.

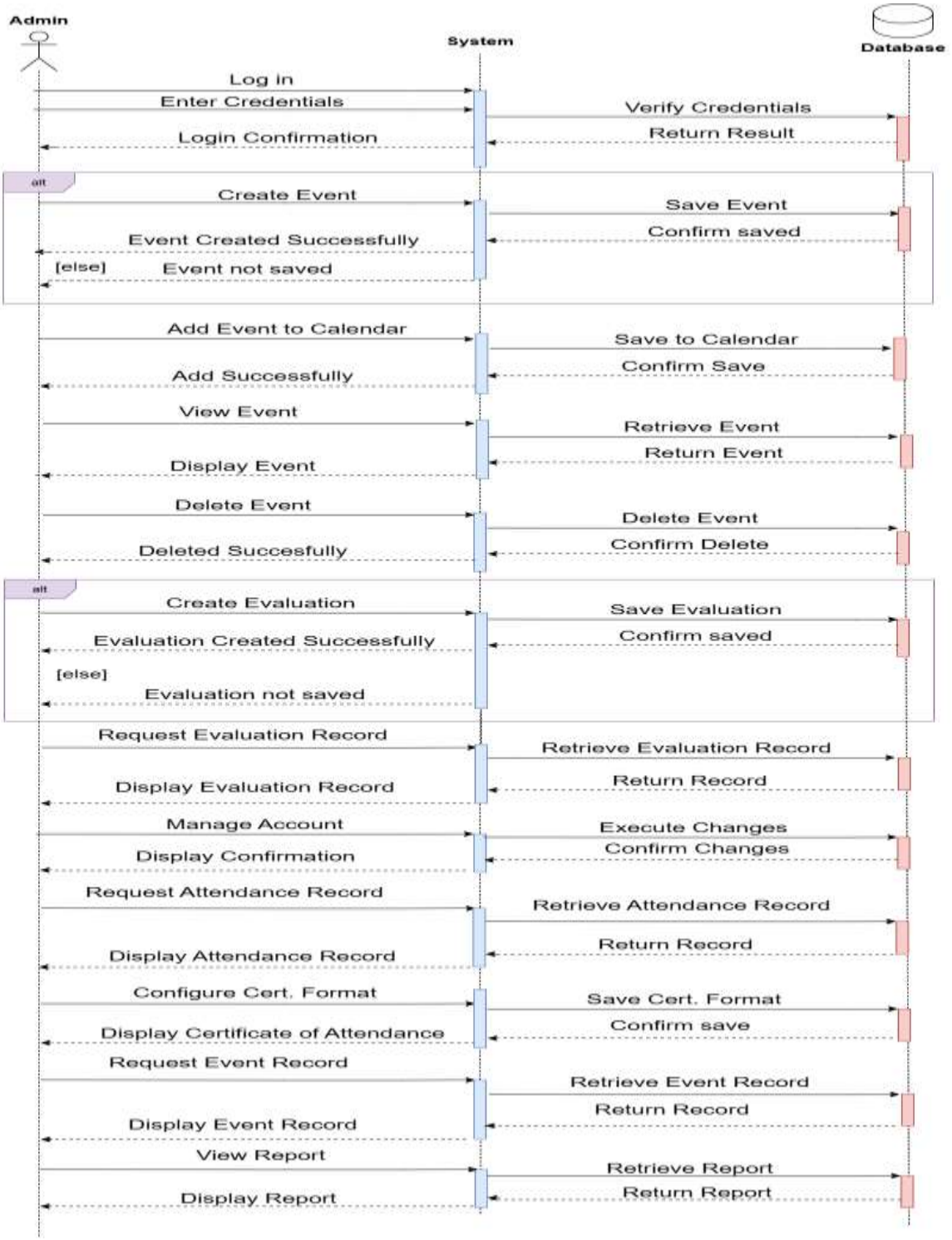


Figure 12. Sequence Diagram-Admin

Figure 12 illustrates the sequence of interactions between the administrator, the system, and the database. It begins with login authentication, followed by administrative tasks such as creating, viewing, and deleting events. The administrator can configure evaluations, manage accounts, and request evaluation, attendance, and event records. Each action triggers system-level operations that interact with the database to save, retrieve, or confirm changes. The admin also configures certificate formats and views generated reports, ensuring end-to-end control over event documentation and participation tracking.

2.7 Entity Relationship Diagram

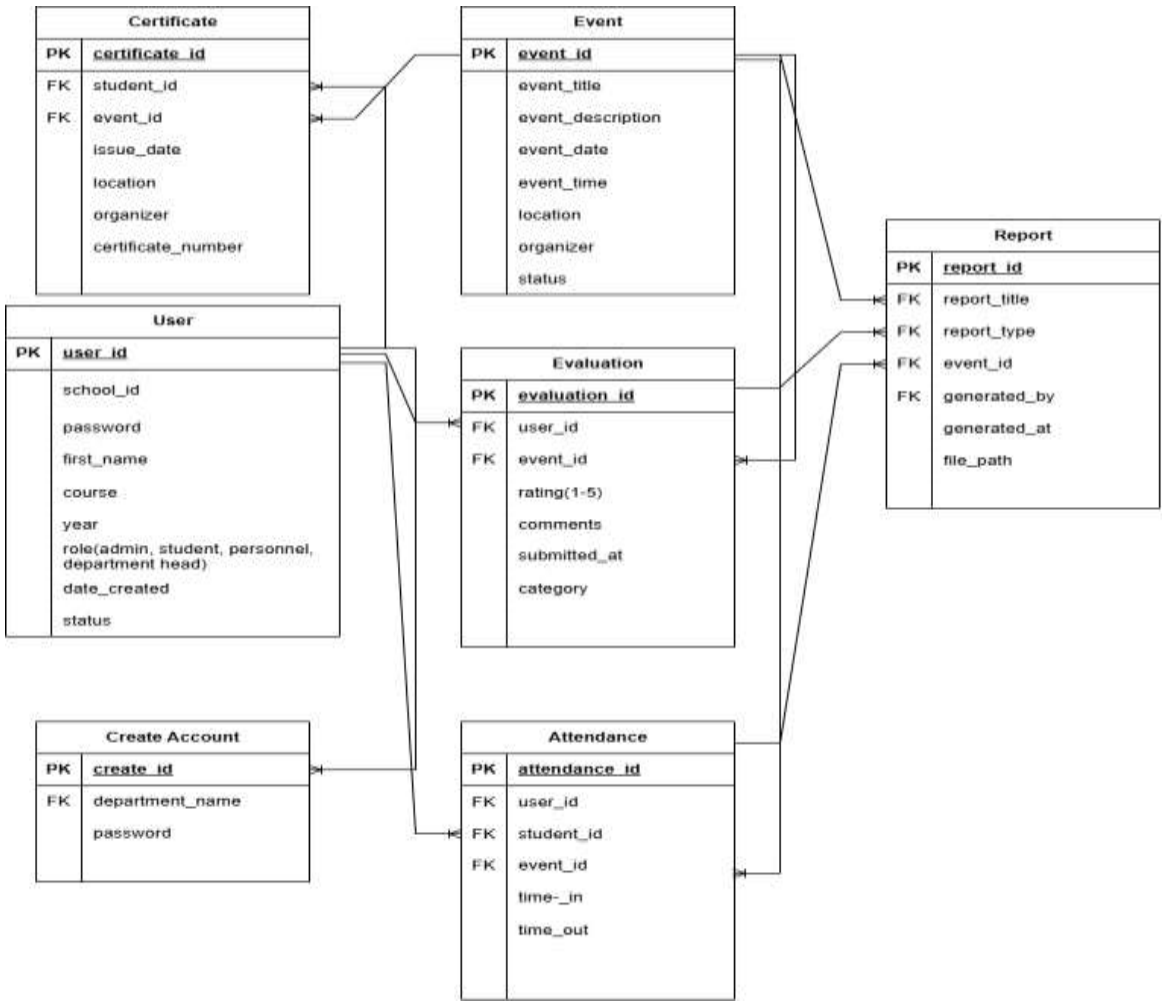


Figure 13. Entity Relationship Diagram

Figure 13 shows the ERD of the Event Evaluation and Attendance System, featuring seven entities: User, Event, Evaluation, Attendance, Certificate, Report, and Create Account. It outlines how user data, event details, feedback, attendance logs, certificates, and reports are interconnected to support secure account creation, event tracking, and documentation.

2.8 User Interface Design



Figure 14. Login Interface of the SMCC Information System-Users

Figure 14 interface illustrates a login form with fields for ID number and password, links for password recovery and registration, and a yellow “Sign In” button. The background shows an aerial view of the campus.

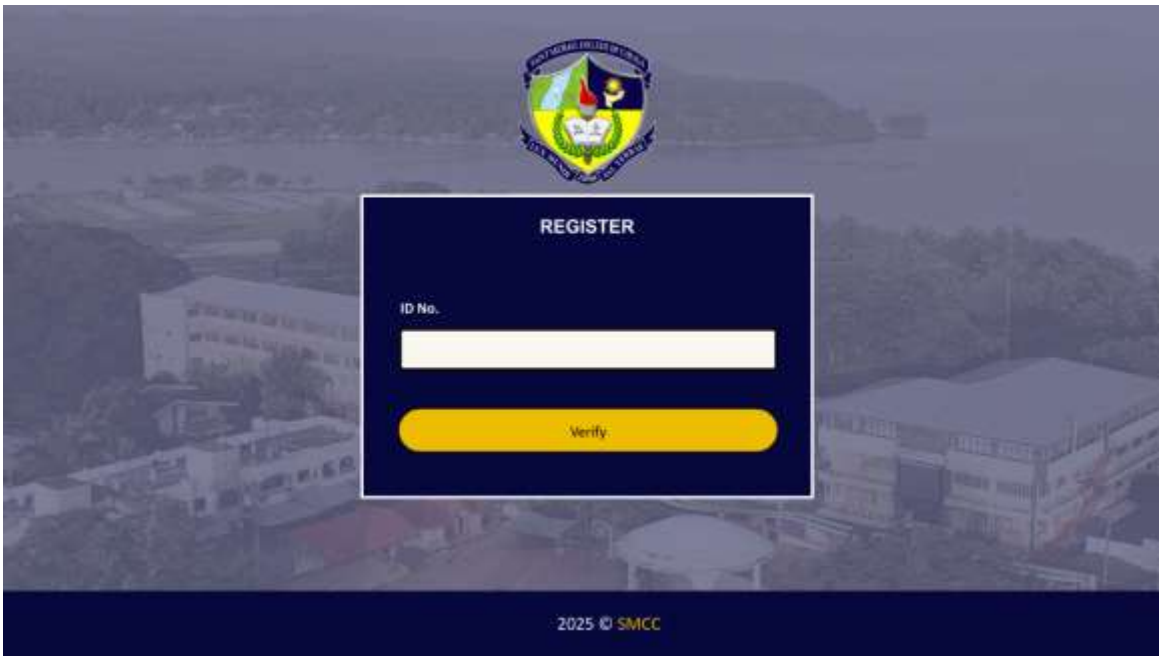


Figure 15. Registration Interface-Users

Figure 15 shows the main entry point of the web-based system developed for Saint Michael College of Caraga. On the left side is the official logo, which reflects the institution's identity and values. On the right side, there are welcome messages "Welcome Michaelinian!" and given two options: "LOG IN" and "REGISTER." This interface is simple and accessible, allowing both students and personnel to access the system.

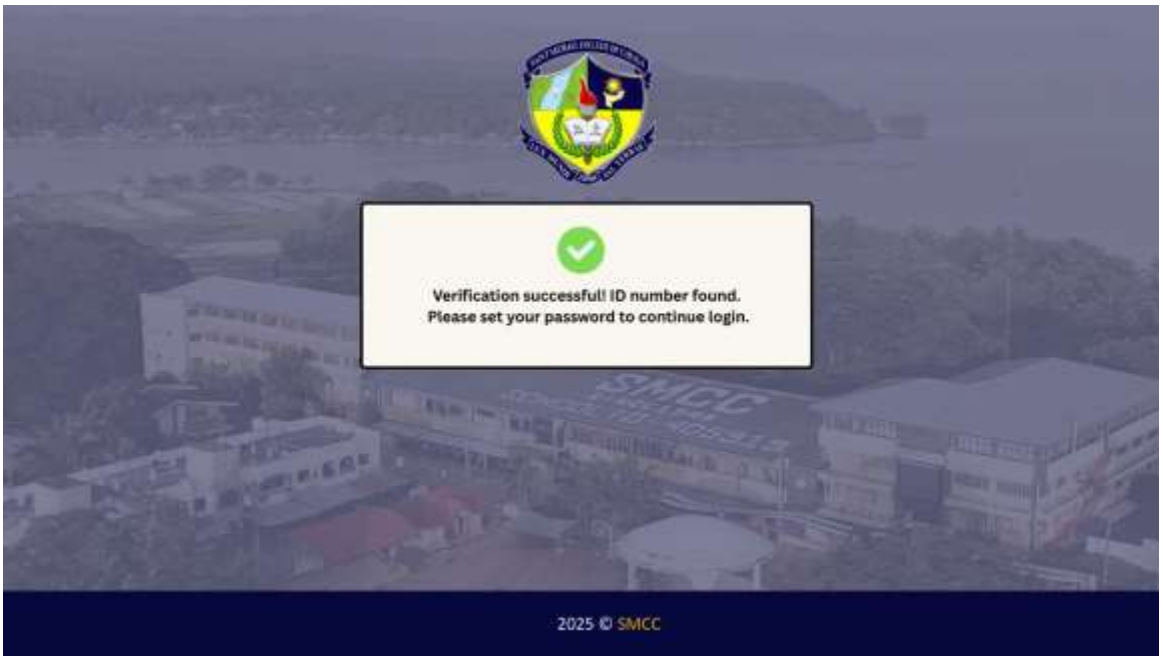


Figure 16. Login Verification Interface-Users

Figure 16 illustrates the login verification interface of Saint Michael College of Caraga (SMCC) for the year 2025. Centered at the top is the official SMCC logo, reinforcing institutional identity. Below it, a message box with a green checkmark icon confirms successful ID number verification, prompting the user to set a password to proceed with login.

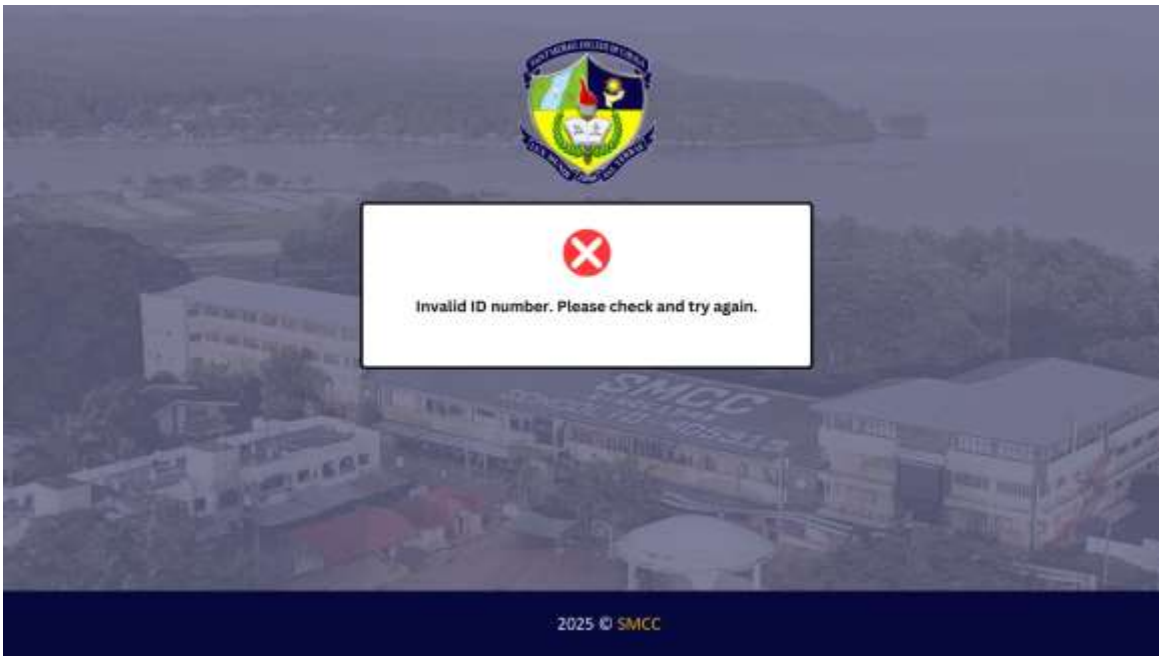


Figure 17. Invalid ID Notification Interface-Users

Figure 17 illustrates an error notification interface from Saint Michael College of Caraga (SMCC) for the year 2025. Positioned at the top center is the official SMCC logo, reinforcing institutional identity. In the middle of the screen, a dialog box with a red “X” icon displays the message: “Invalid ID number. Please check and try again,” indicating that the system failed to recognize the user’s input.



Figure 18. Registration Form Interface-Users

Figure 18 showcases the registration form interface of Saint Michael College of Caraga (SMCC) for the year 2025. Set against a campus-themed background, the form includes fields for setting and confirming a password, entering a Gsuite account email (used for certificate delivery), selecting a security question, and providing a corresponding answer. A yellow “Register” button is positioned at the bottom, accompanied by a reminder note emphasizing the importance of remembering the security answer for future password recovery.

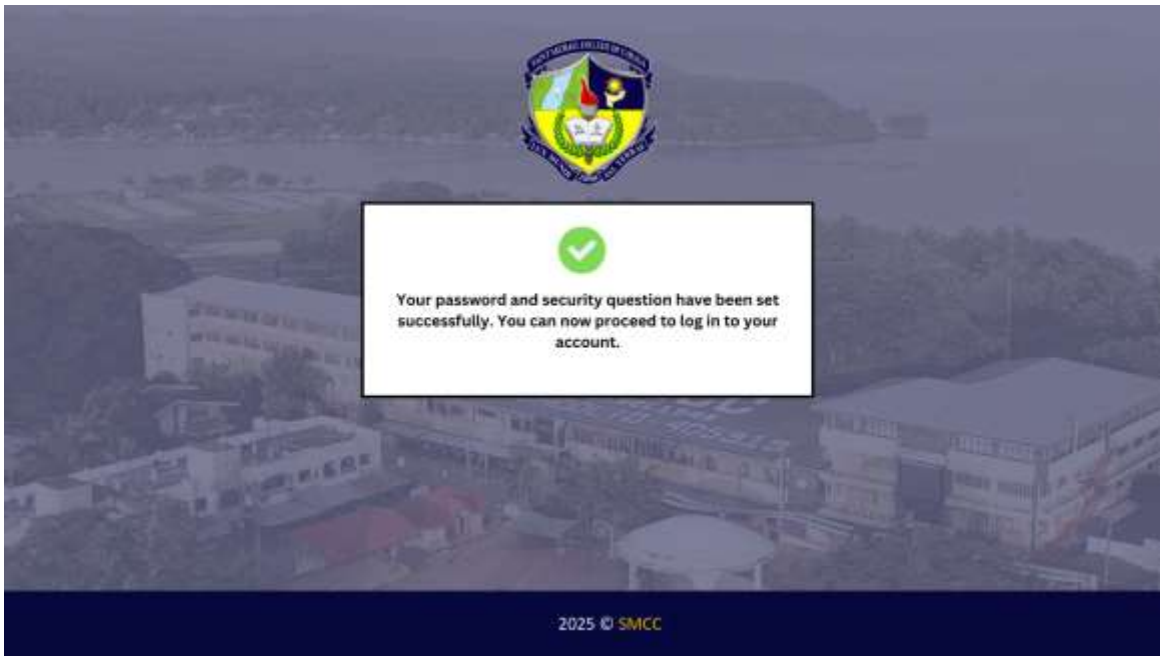


Figure 19. Security Setup Confirmation Interface-Users

Figure 19 illustrates a confirmation interface from Saint Michael College of Caraga (SMCC) for the year 2025, indicating successful completion of account security setup. At the top center is the official SMCC logo, followed by a checkmark icon and a message box stating: “Your password and security question have been set successfully. You can now proceed to log in to your account.”

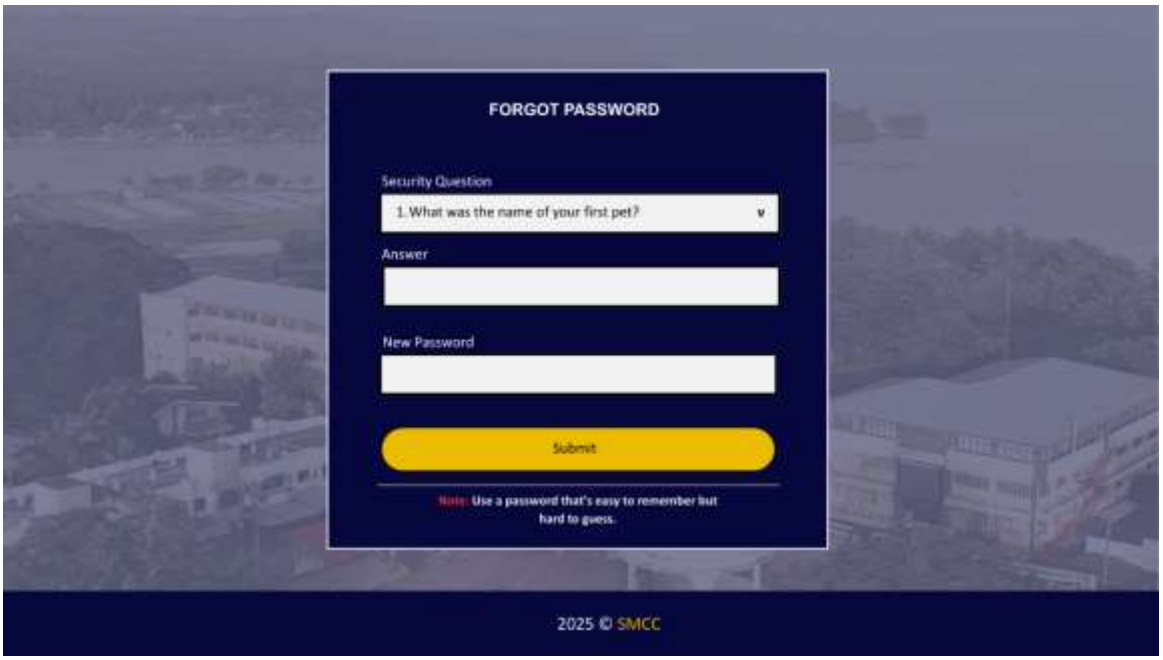


Figure 20. Forgot Password Interface-Users

Figure 20 The figure presents the password recovery interface of Saint Michael College of Caraga (SMCC) for the year 2025. The form includes a dropdown menu for selecting a security question, a field to input the corresponding answer, and a field to enter a new password. A yellow “Submit” button is positioned below the input fields. A red advisory note reminds users to choose a password that is easy to remember but difficult to guess.

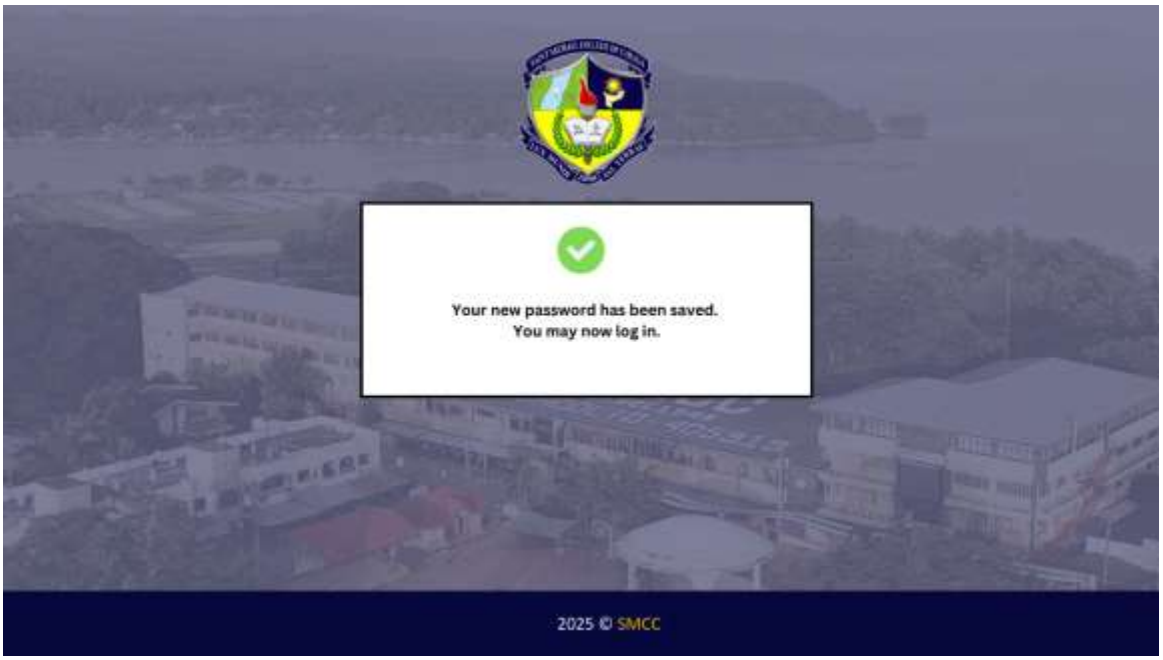


Figure 21. Password Update Confirmation Interface-Users

Figure 21 illustrates a successful password update. Centered at the top is the official SMCC logo, followed by a white dialog box featuring a green checkmark icon and the message: “Your new password has been saved. You may now log in.”

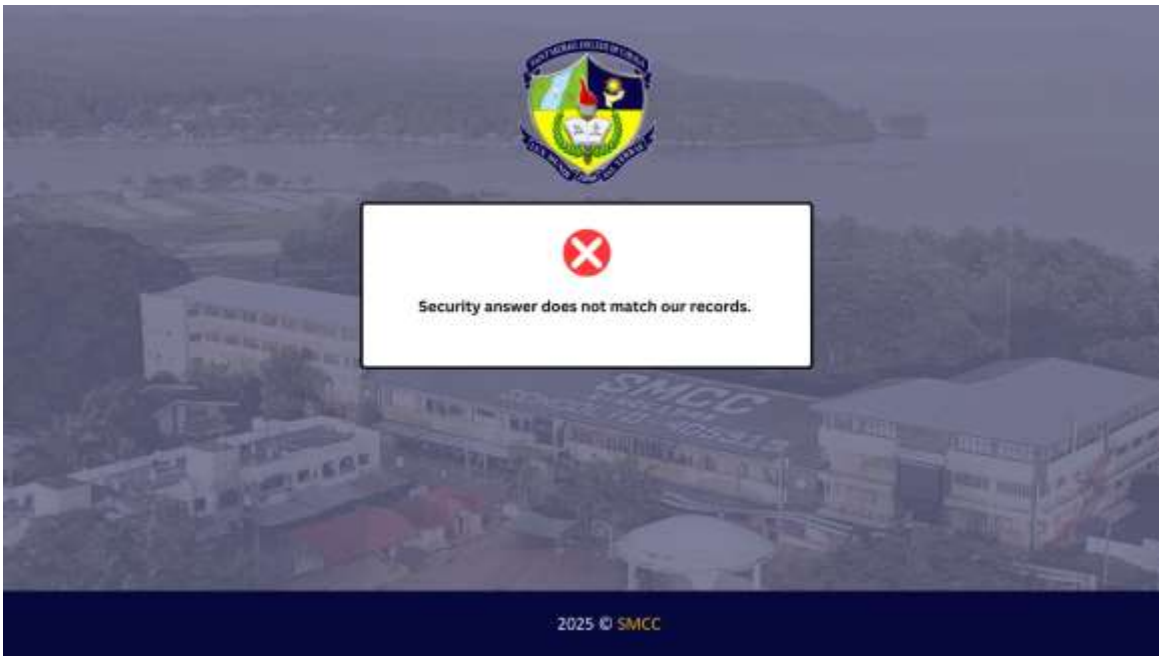


Figure 22. Security Answer Error Interface-Users

Figure 22 displays a pop-up dialog box displaying a red “X” icon and the message: “Security answer does not match our records.” This indicates a failed attempt to verify the user’s identity using a security question. The background features an aerial view of the SMCC campus, with visible buildings and surrounding greenery.



Figure 23. Student Profile Interface-Student/Personnel

Figure 23 shows a digital student profile interface. The layout includes a left-side navigation menu with options: Profile, Event, History, Change Password, and Sign Out. The main section displays student information such as ID number, first and last name, course (BSIT), and year level (3rd Year), along with a placeholder for uploading a profile picture.



Figure 24. Event Calendar Interface-Student/Personnel

Figure 24 displays a digital event calendar interface featuring multiple entries of the same scheduled activity titled “CCIS Seminar.” Each instance is set for October 8 from 1:00 PM to 5:00 PM at the AVR (Audio-Visual Room). The layout includes a left-side navigation menu with options such as Profile, Event, History, Change Password, and Sign Out. A top navigation bar provides access to sections labeled CCIS, ESG, OSAS, GUIDANCE, and Calendar of Events. Each event is presented in a green-bordered box, indicating repeated listings of identical seminar details. The background includes a faint aerial view of a campus.

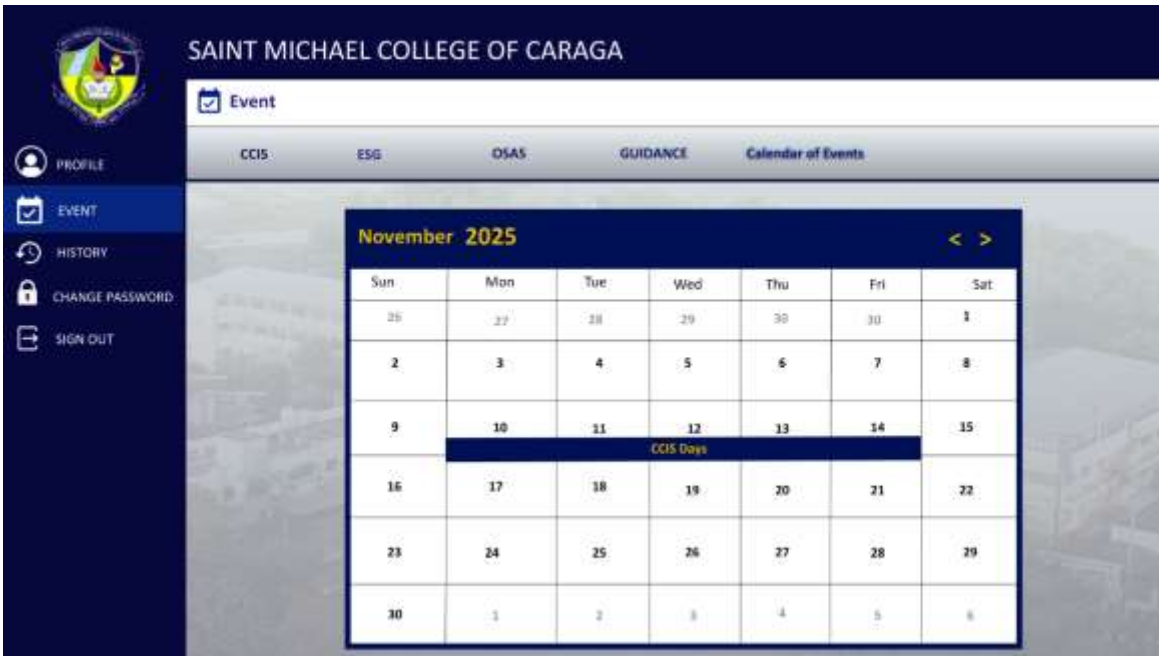


Figure 25. Calendar of Events Interface-Student/Personnel

Figure 25 shows a digital calendar interface highlighting an event labeled “CCIS Days,” scheduled from November 10 to November 14. The layout includes a left-side navigation menu with options such as Profile, Event, History, Change Password, and Sign Out. A top navigation bar provides access to sections labeled CCIS, ESG, OSAS, GUIDANCE, and Calendar of Events. The calendar is set to the month of November and visually marks the multi-day event.

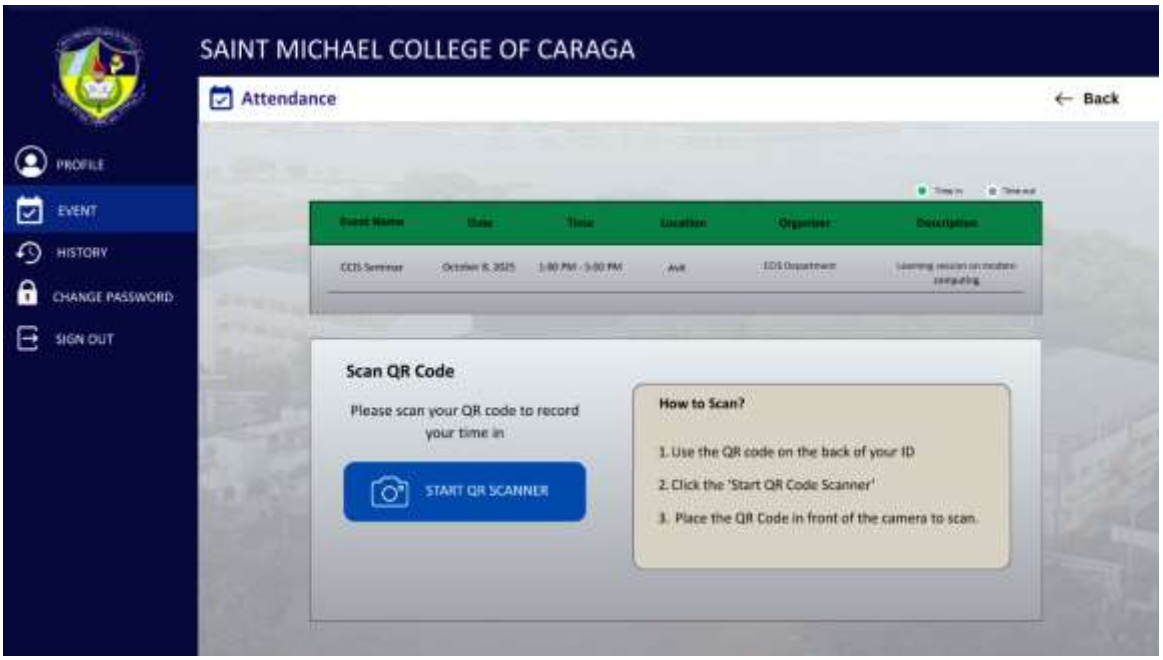


Figure 26. QR Code Attendance Logging Interface-Students/Personnel

Figure 26 shows users to scan the QR code located on the back of their ID to record attendance. By clicking the “Start QR Scanner” button and positioning the code in front of the camera, the system captures the time-in entry efficiently. Clear instructions guide the scanning process to ensure accurate and secure attendance logging.

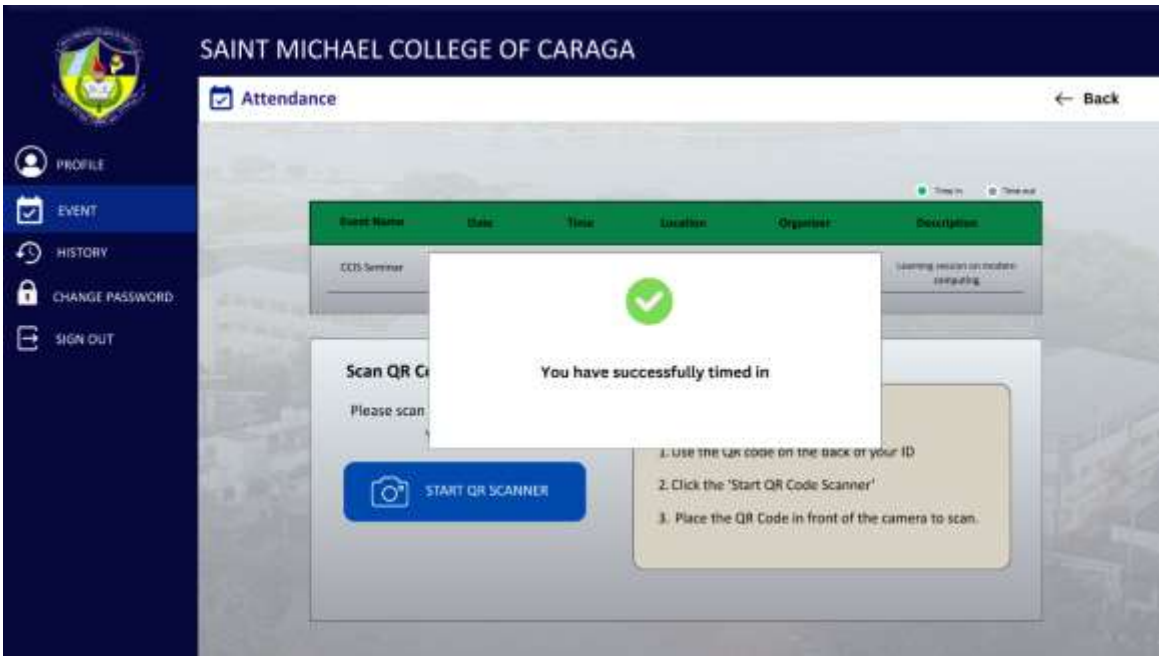


Figure 27. QR Code Check-In Confirmation Interface-Students/Personnel

Figure 27 displays successful attendance logging via QR code. A green checkmark and message indicate that the user has successfully timed in for the event. The interface reinforces the use of QR code scanning for efficient and secure attendance tracking, with clear instructions and a “Start QR Scanner” button facilitating the process.

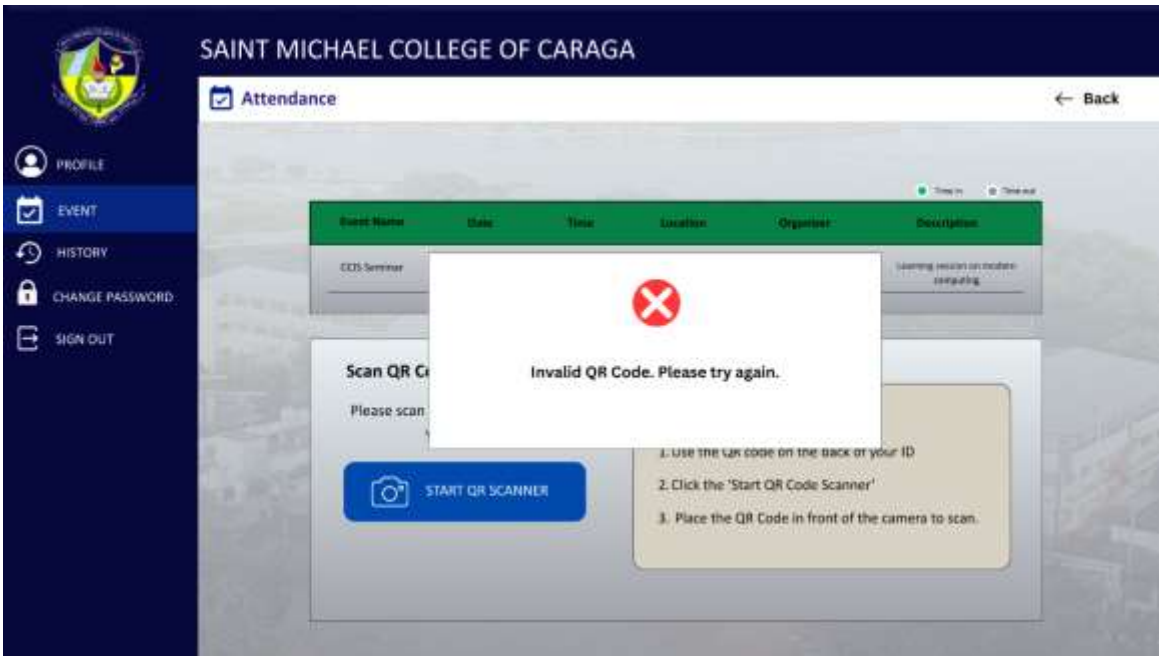


Figure 28. Invalid QR Code Notification Interface-Students/Personnel

Figure 28 shows an error message indicating a failed attempt to log attendance due to an unrecognized QR code. A red “X” icon accompanies the message: “Invalid QR Code. Please try again.” Users are instructed to scan the QR code located on the back of their ID by clicking the “Start QR Scanner” button and positioning the code in front of the camera. This prompt ensures proper attendance logging through secure and accurate QR code validation.

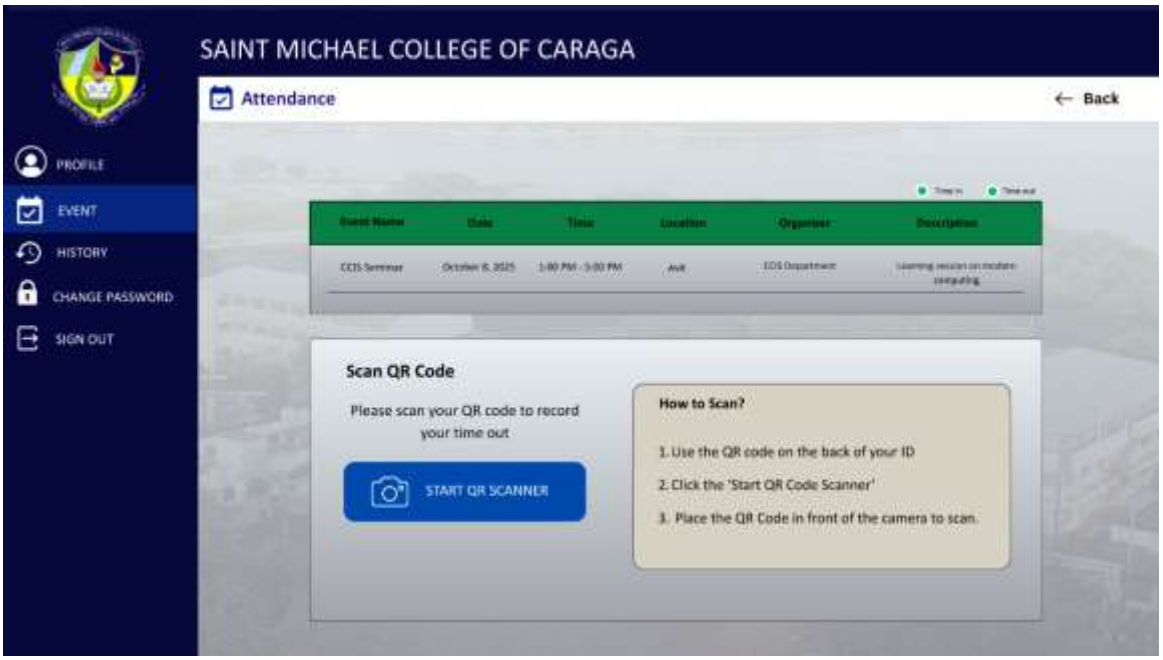


Figure 29. QR Code Time-Out Logging Interface-Student/Personnel

Figure 29 shows the recording of time-out attendance using a QR code. Users are prompted to scan the QR code located on the back of their ID by clicking the “Start QR Scanner” button and positioning the code in front of the camera. Clear step-by-step instructions guide the process, ensuring accurate and secure logging of departure time from the event.

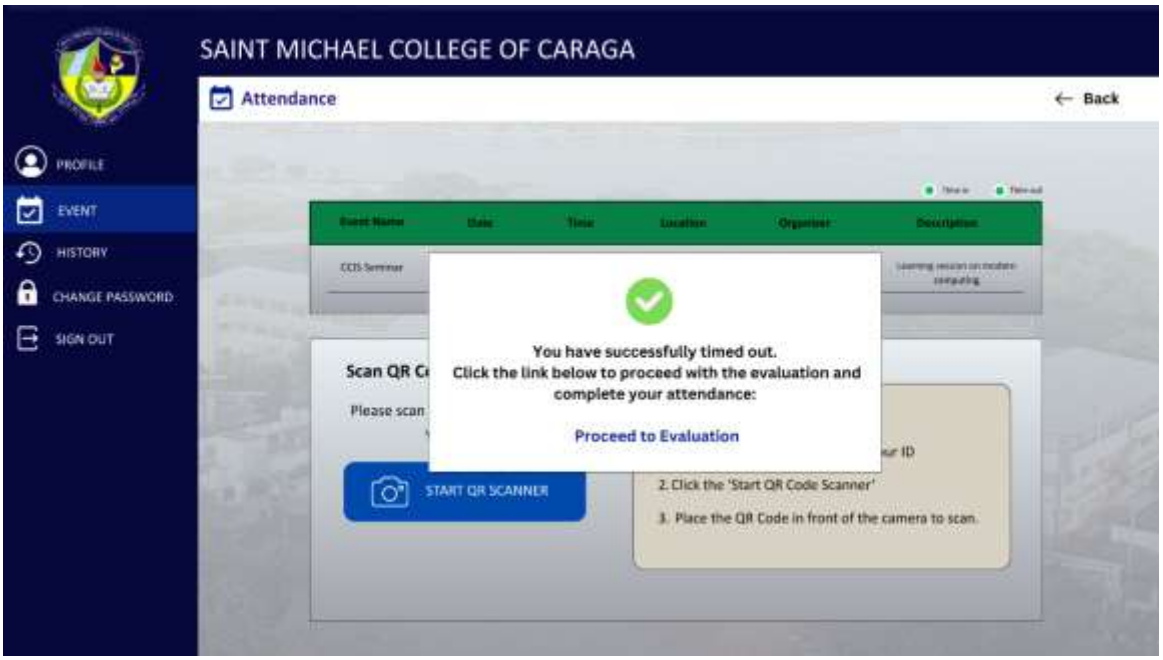


Figure 30. Time-Out Confirmation and Evaluation Prompt Interface-Student/Personnel

Figure 30 displays the user profile interface for personnel. It has a sidebar navigation with options: Profile, Event, Attendance, Change Password, and Sign Out. The main section shows user details, including ID number and full name. There is also a placeholder for a profile picture.

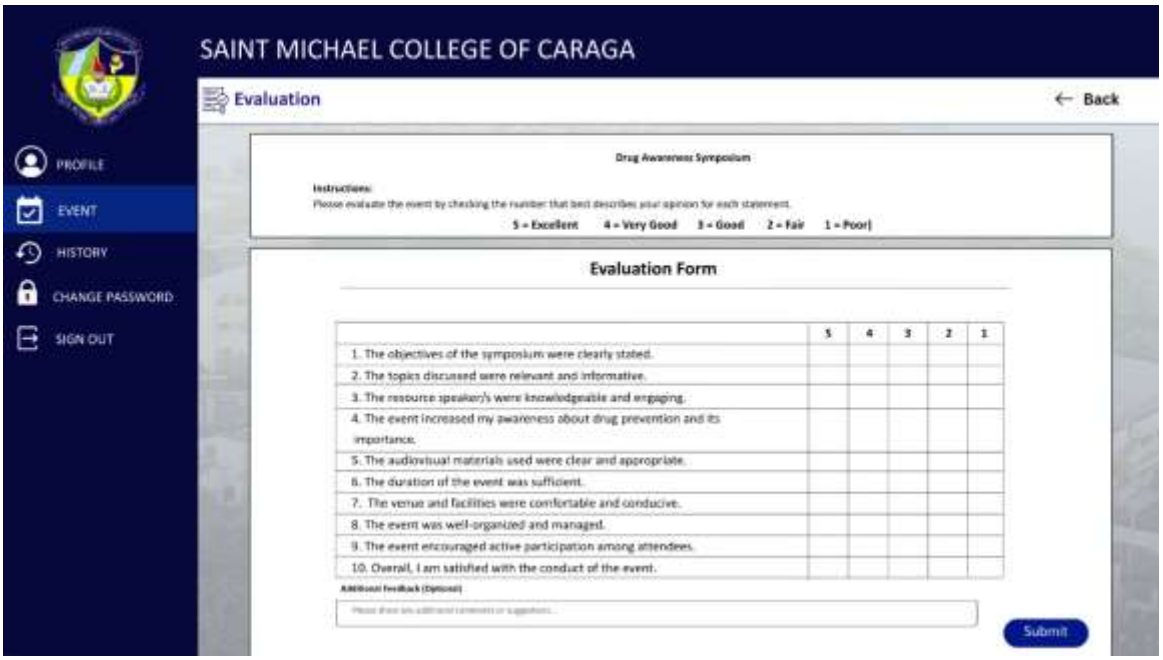


Figure 31. Time-Out Confirmation and Evaluation Access Interface-Student/Personnel

Figure 31 shows that the interface has the same layout and features as the student version (see Figure 21). It displays event details such as the name, date, time, location, organizer, and description. Below the event information is a “Scan QR Code” section with instructions and a button to start the scanner. Personnel can use their ID’s QR code to confirm attendance. Keeping the same design for both students and personnel makes the system easier to use and understand

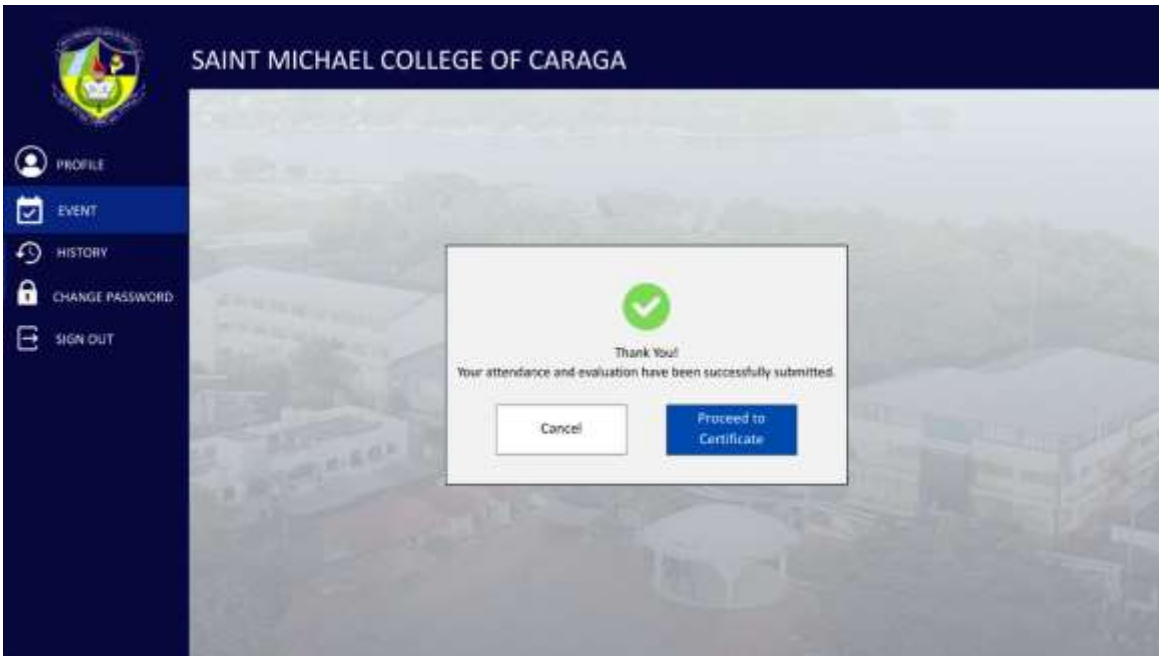


Figure 32. Attendance and Evaluation Submission Confirmation Interface-Student/Personnel

Figure 32 shows the Attendance interface used by personnel. The layout and design are the same as the department head interface, to keep users familiar with the system. This interface displays event details such as the name, date, location, and total number of attendees within their department. Below is a list of academic programs, each with a link to view the attendance records. It helps personnel monitor participation across various programs in their department.



Figure 33. Certificate of Attendance Interface-Student/Personnel

Figure 33 displays a digital certificate of attendance awarded to Juan Dela Cruz for participating in the “Drug Awareness Symposium” held on October 10 at the Quadrangle. The event was organized by the Office of Student Affairs and acknowledges the participant’s active involvement and commitment. The certificate is shown within a webpage interface featuring a sidebar with navigation options: Profile, Event, History, Change Password, and Sign Out. Action buttons labeled “Send to Gsuite” and “Download PDF Report” are available for certificate management.



Figure 34. Attendance History Interface

Figure 34 shows a digital attendance history interface listing multiple records for an event titled “CCIS Seminar,” held on October 8 at 1:00 PM. Each entry includes action links labeled “Evaluation” and “Certificate,” allowing users to review feedback and access their attendance certificate. The left sidebar features navigation options: Profile, Event, History (highlighted), Change Password, and Sign Out.

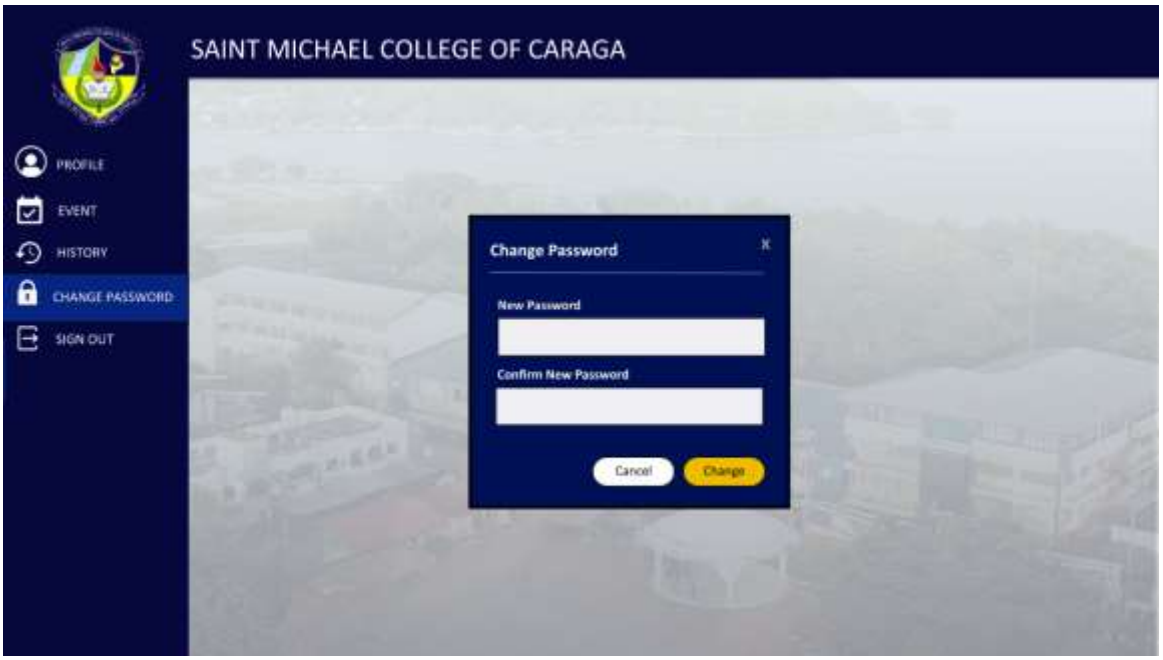


Figure 35. Password Change Interface-Student/Personnel

Figure 35 shows a password change interface within a user portal. A modal window titled "Change Password" contains two input fields labeled "New Password" and "Confirm New Password," along with action buttons: "Cancel" and "Change." The left sidebar includes navigation options such as Profile, Event, History, Change Password (highlighted), and Sign Out.



Figure 36. Attendance Record Interface-Personnel

Figure 36 shows the dashboard, including total events, total attendees, and attendance rate, along with monthly changes. A pie chart and table display evaluation and attendance data by academic program. This will help department heads to monitor event performance and participation within their department.



Figure 37. Program-Based Attendance Record Interface-Personnel

Figure 37 shows a digital attendance record interface displaying event participation data segmented by academic program. The listed event is “CCIS Seminar,” held on October 15 at the Quadrangle, with a total of 120 attendees. A table outlines the participating programs: Bachelor of Science in Information Technology, Bachelor of Librarian in Information Science, Bachelor of Science in Computer Science, and Bachelor of Science in Information System. Each program row includes a “View full record” action link. A dropdown menu labeled “Year Level” is positioned at the top right, allowing users to filter attendance data by academic year.



Figure 38. Attendance List Interface-Personnel

Figure 38 displays a digital attendance list for the “CCIS Seminar” held on October 8, 2023, from 8:00 AM to 5:00 PM at the SMCC AVR. The table includes twenty identical entries for a student named Juan dela Cruz, enrolled in the Bachelor of Science in Information Technology (BSIT) program and currently in their 4th year.



Figure 39. Event Dashboard Overview Interface -Department Head

Figure 39 displays a digital dashboard interface summarizing event-related metrics for Saint Michael College of Caraga. Key indicators include an attendance rate of 87%, an average evaluation score of 50%, a total of 5 events conducted, and 335 total attendees. A pie chart titled “Event Evaluation and Attendance” visualizes the percentage distribution across four academic programs: BSIT (33.8%), BSIS (19.7%), BSCS (21.1%), and BLIS (25.4%)—with identical values for both evaluation and attendance. A date filter and “Report” button are positioned at the top, enabling dynamic data review.

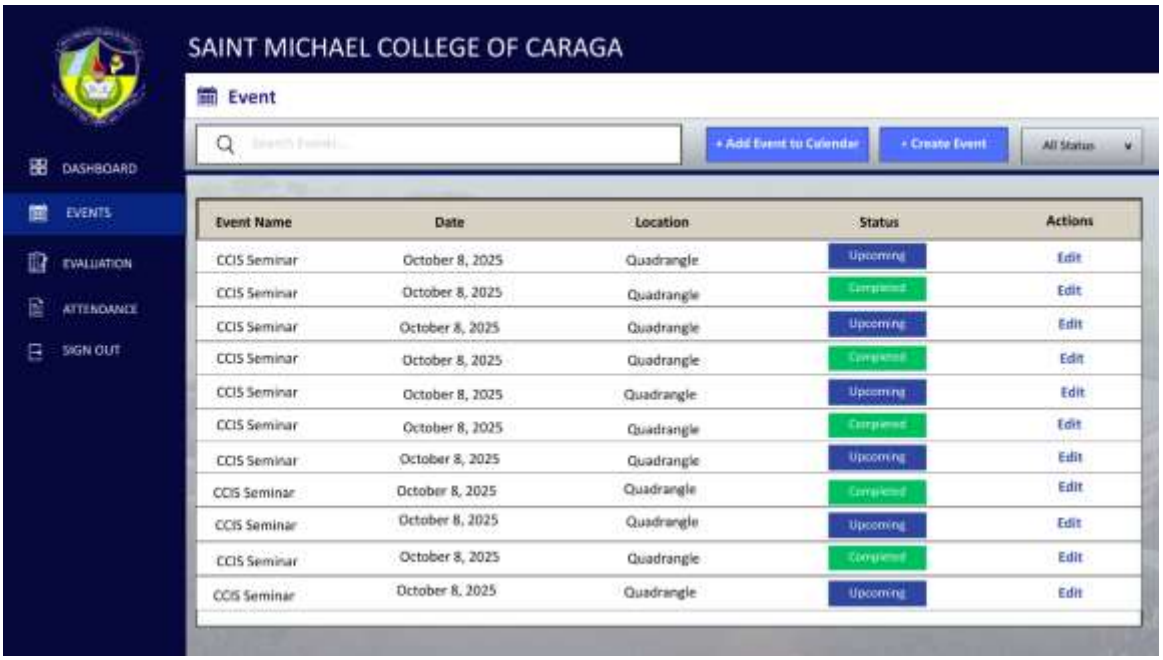


Figure 40. Event Management Interface-Department Head/Admin

Figure 40 shows a digital event management interface listing multiple entries for the "CCIS Seminar" scheduled on October 8, 2025, at the Quadrangle. Each row displays the event name, date, location, status (either "Upcoming" or "Completed"), and an "Edit" action link. Above the table are controls for adding or creating events, filtering by status, and searching by event name.



Figure 41. Event Configuration Interface-Department Head/Admin

Figure 41 shows an event configuration interface for the “CCIS Seminar” scheduled on October 8, 2025, at the Quadrangle. The interface allows administrators to enable or disable attendance logging options for “Time in” and “Time out” using toggle indicators: green for enabled, grey for disabled, and red for deactivated. Two action buttons “Cancel” and “Save” are positioned at the bottom for managing changes.



Figure 42. Calendar-Based Event Scheduling Interface-Department Head/Admin

Figure 42 displays a digital calendar interface set to November 2025, designed for plotting and managing institutional events. The calendar grid displays dates from late October through early December, with navigation controls for month switching. Action buttons labeled “Add Event to Calendar” and “Manage Plotted Events” allow users to schedule and organize activities.



Figure 43. Event Scheduling and Calendar Plotting Interface-Department Head/Admin

Figure 43 illustrates a calendar-based scheduling interface highlighting the event “CCIS Days,” plotted from Tuesday the 11th to Friday the 14th. A pop-up form is active, showing fields for “Event Title,” “Start Date,” and “End Date,” with “CCIS Days” pre-filled. A “Save” button allows users to confirm the event entry. The top of the calendar includes controls for adding events and managing plotted entries.

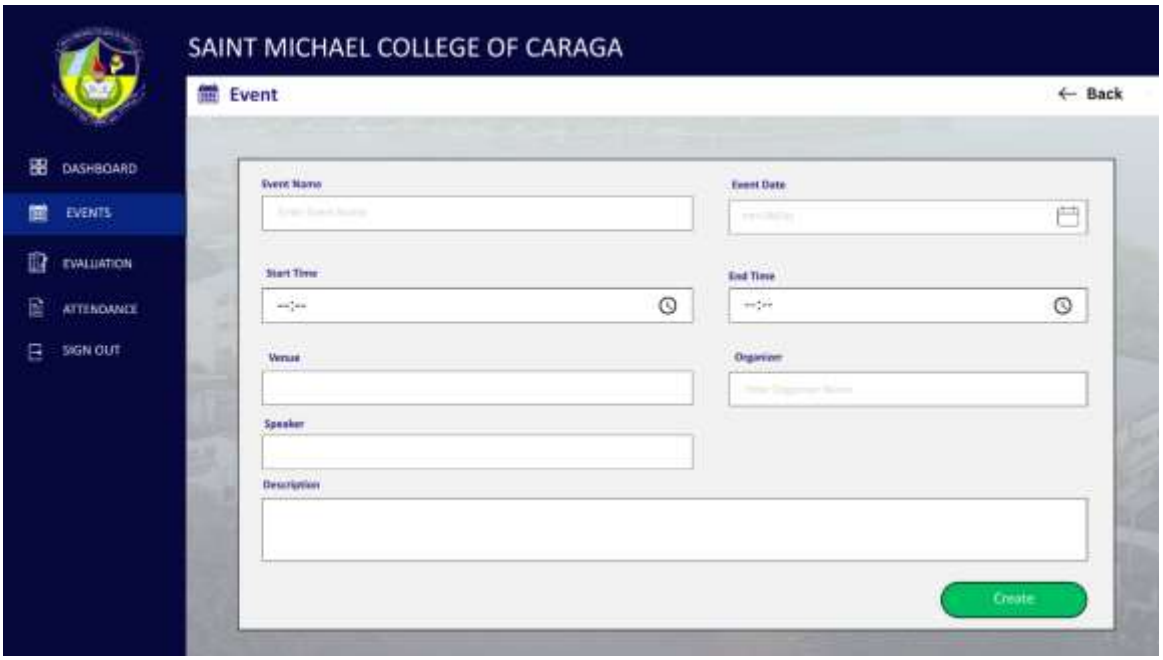


Figure 44. Event Creation Form Interface-Department Head/Admin

Figure 44 shows a structured event creation form designed for scheduling academic activities. Input fields include Event Name, Event Date (with calendar selector), Start Time, End Time (both with clock icons), Venue, Organizer, Speaker, and Description. A green “Create” button is positioned at the bottom right to finalize the entry.

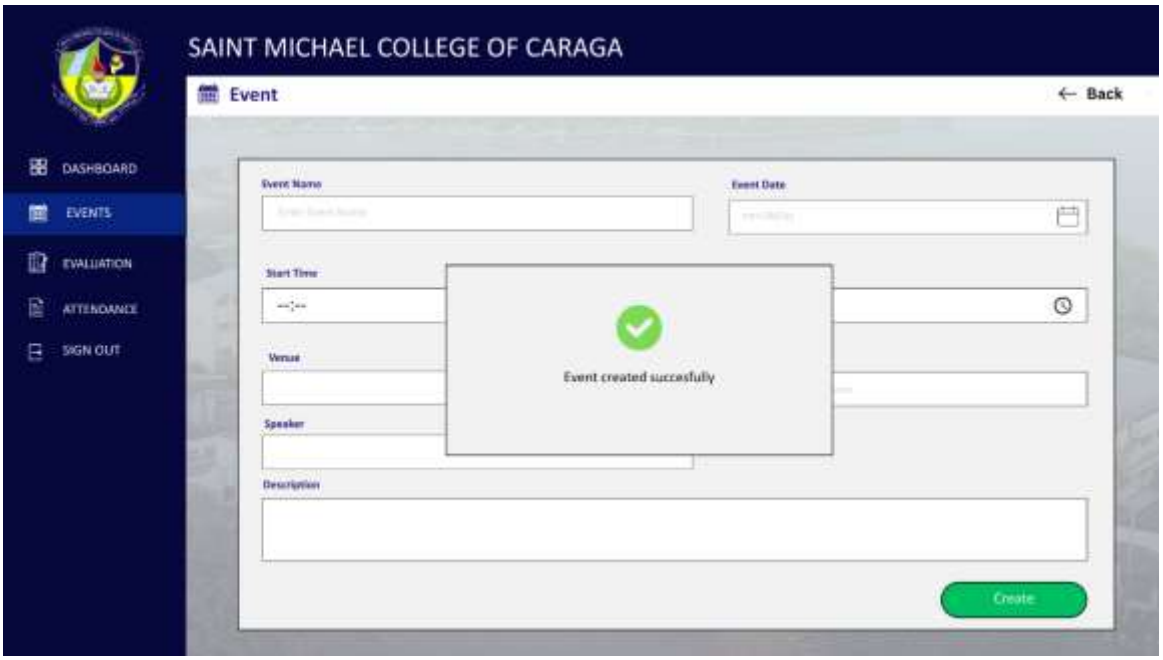


Figure 45. Event Creation Confirmation Interface-Department Head/Admin

Figure 45 displays a confirmation overlay indicating successful creation of an event titled “CCIS Days.” A green checkmark icon accompanies the message “Event created successfully.” The underlying form includes fields for Event Name, Event Date, Start Time, Venue, Speaker, and Description.

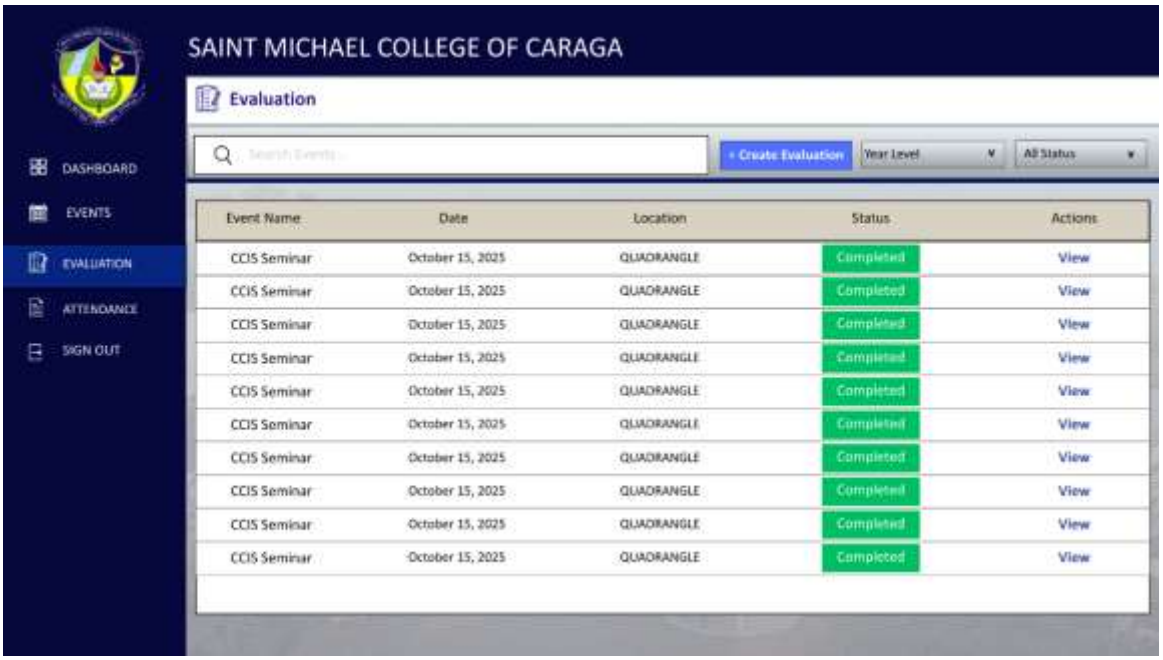


Figure 46. Event Evaluation Management Interface-Department Head/Admin

Figure 46 shows an event evaluation management interface listing multiple entries for the “CCIS Seminar” held on October 15, 2025, at the Quadrangle. Each row includes the event name, date, location, status marked as “Completed,” and a “View” action link. Above the table are controls for creating evaluations, filtering by year level and status, and searching by event name.



Figure 47. Evaluation Form Creation Interface-Department Head/Admin

Figure 47 displays a structured evaluation form interface used for creating feedback instruments within the institutional dashboard. Users can copy an existing form or input a new Evaluation Title and Instructions. The form includes “Category 1: Speaker Evaluation” with editable questions and rating columns: Excellent, Very Good, Good, Fair, Poor, and Edit. Each rating field is initialized to zero. Additional categories and questions can be added via dedicated links. Action buttons “Cancel” and “Save” are positioned at the bottom.



Figure 48. Evaluation Creation Confirmation Interface-Department Head/Admin

Figure 48 illustrates a confirmation overlay indicating successful creation of an evaluation form. A green checkmark icon accompanies the message “Evaluation created successfully.” The underlying form includes fields for Evaluation Title and Instructions, along with a category labeled “Speaker Evaluation.” Each question row includes rating columns for “Fair” and “Poor,” initialized to zero, and an “Edit” link. Users can add more questions and categories using dedicated controls.

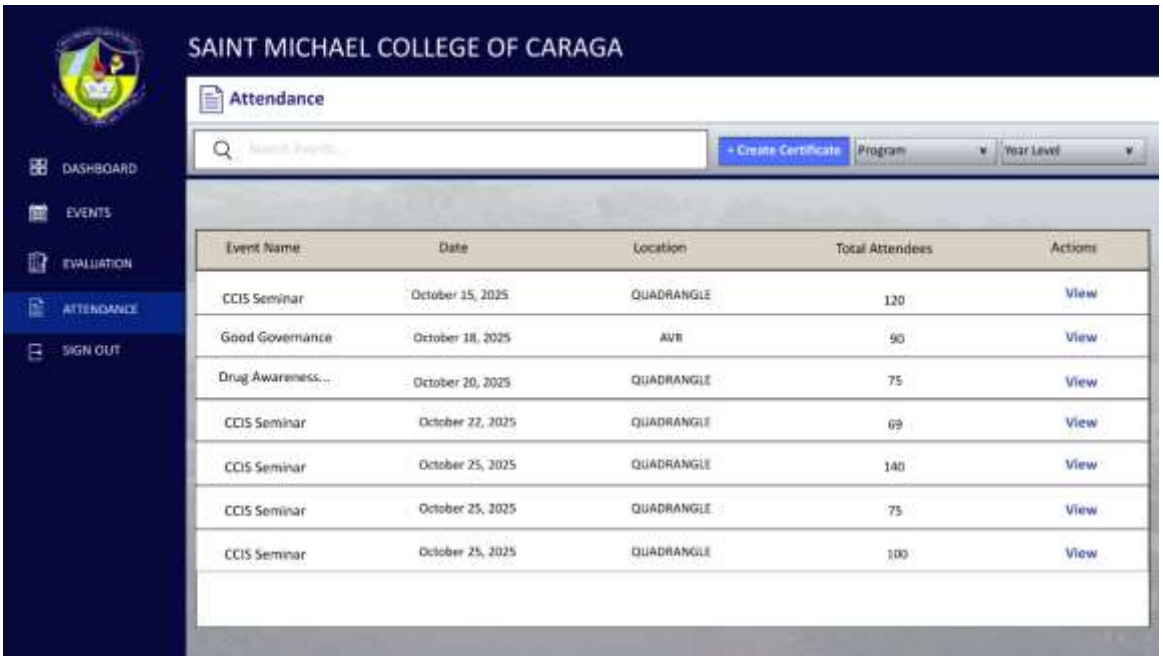


Figure 49. Attendance Overview and Certificate Management Interface-Department Head/Admin

Figure 49 presents a digital attendance tracking interface listing multiple institutional events, including “CCIS Seminar,” “Good Governance,” and “Drug Awareness,” with respective dates, venues (Quadrangle and AVR), and total attendee counts ranging from 69 to 140. Each row includes a “View” action link for accessing detailed records. Above the table are controls for certificate creation, program selection, and year-level filtering, along with a search bar for event lookup.



Figure 50. Student and Personnel Attendance Management Interface-Department Head/Admin

Figure 50 displays a digital attendance management interface listing six identical entries for the “CCIS Seminar” held on October 15, 2025, at the Quadrangle, with a total of 120 attendees per entry. The Students tab is active, indicating that the records pertain to student participants. Each row includes a “View” action link for accessing detailed attendance data.



Figure 51. Attendance Report Interface-Department Head/Admin

Figure 51 displays a digital attendance report interface for the “CCIS Seminar” held on October 8, 2023, from 8:00 AM to 5:00 PM at the SMCC AVR. The table lists twenty identical entries for a student named Juan dela Cruz, enrolled in the Bachelor of Science in Information Technology (BSIT) program and in their 4th year.



Figure 52. Certificate Generation Interface for Event Attendance-Department Head/Admin

Figure 52 shows a certificate generation interface for the “Drug Awareness Symposium” held on October 10, 2025, at the Saint Michael College of Caraga Quadrangle. The certificate template includes decorative borders and placeholder lines for recipient names. The text acknowledges active participation and commitment, issued by the Office of Student Affairs. A “Generate” button is located at the bottom right to finalize the certificate.

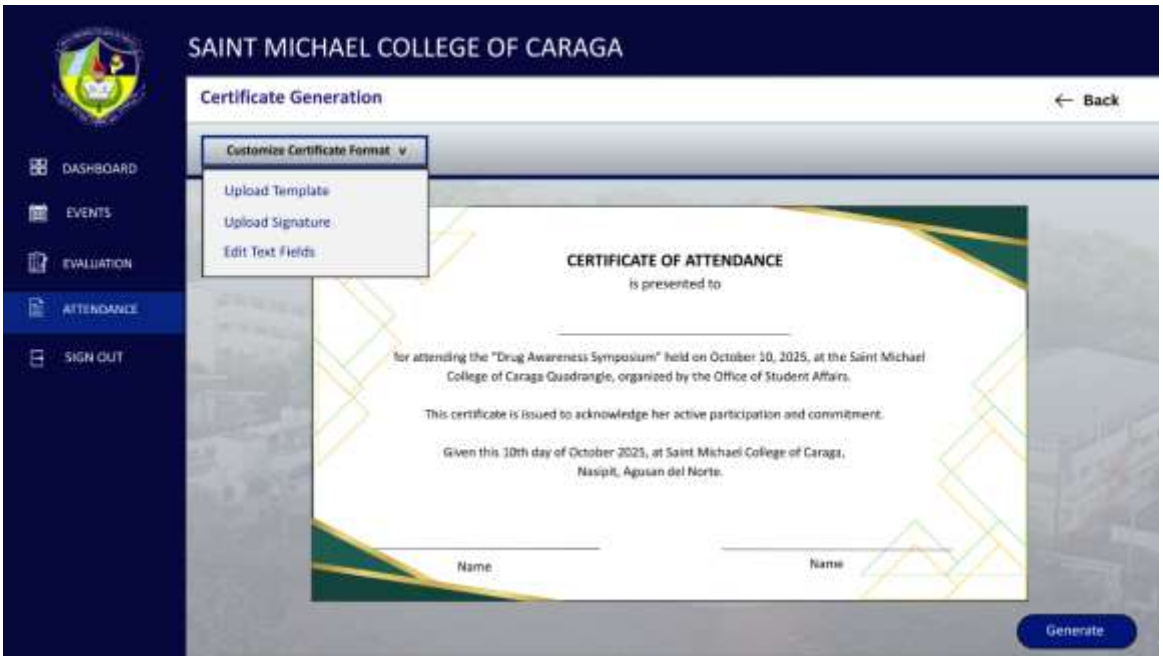


Figure 53. Certificate Customization and Generation Interface-Department Head/Admin

Figure 53 shows a customizable certificate generation interface for the “Drug Awareness Symposium” held on October 10, 2025, at the Saint Michael College of Caraga Quadrangle. The certificate template includes editable fields and a dropdown menu labeled “Customize Certificate Format” with options to Upload Template, Upload Signature, and Edit Text Fields. The certificate acknowledges active participation and commitment, with placeholders for the recipient’s and issuer’s names. A “Generate” button is located at the bottom right.

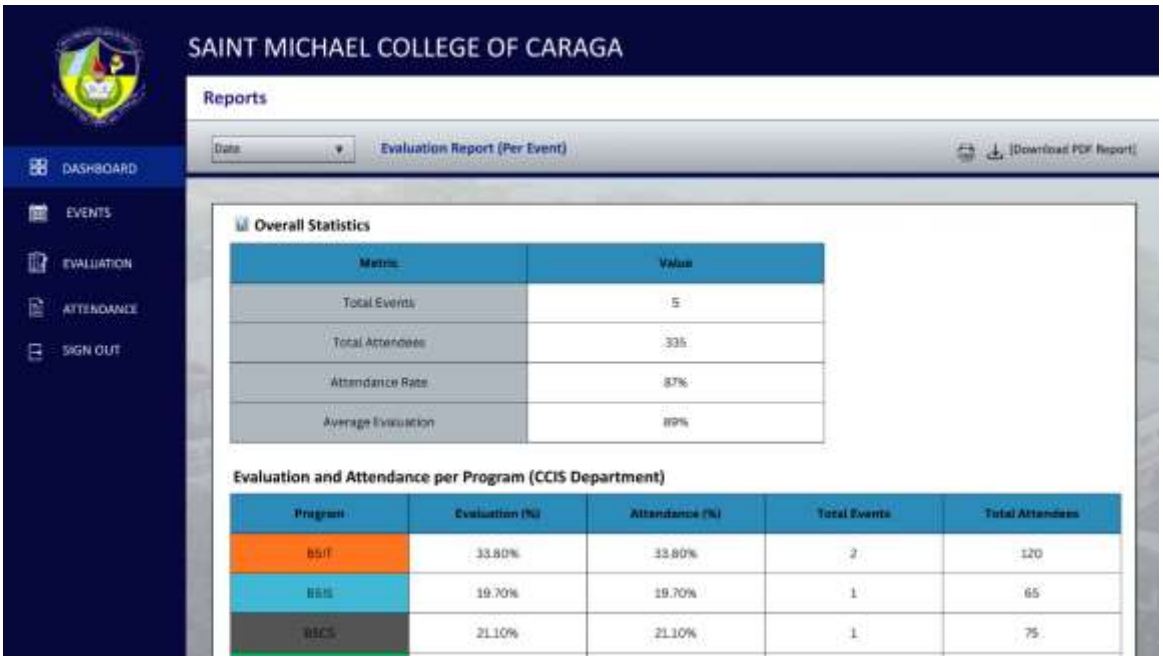


Figure 54. Evaluation Report Summary Interface-Department Head/Admin

Figure 54 displays an “Evaluation Report (Per Event)” interface for the CCIS Department of Saint Michael College of Caraga. The report is divided into two sections. The “Overall Statistics” table summarizes institutional metrics: 5 total events, 335 total attendees, 87% attendance rate, and 89% average evaluation score. The second table, “Evaluation and Attendance per Program,” details program-specific metrics: BSIT with 33.80% evaluation and attendance across 2 events and 120 attendees; BSIS with 19.70% across 1 event and 65 attendees; and BSCS with 21.10% across 1 event and 75 attendees.



Figure 55. Dashboard Interface-Department Head/Admin

Figure 55 presents a detailed evaluation report for the “Drug Awareness Symposium” held on October 12, 2022, at the SMCC Quadrangle. Organized by the Office of Student Affairs and facilitated by Mr. John Dela Cruz, the event recorded 120 respondents with a 96% completion rate. Category-based results include speaker clarity (Excellent: 80%, Very Good: 15%, Good: 5%) and venue accessibility (Excellent: 60%, Very Good: 30%, Good: 10%). Participant comments highlight the speaker’s engagement, event organization, venue comfort, and a suggestion for more interactive activities.



Figure 56. Event Evaluation and Attendance Dashboard Overview-Admin

Figure 56 illustrates a comprehensive dashboard interface summarizing event participation and evaluation metrics across institutional departments. A pie chart titled “Event Evaluation and Attendance” displays equal distribution (11.1%) among nine categories: Elementary, Junior High, Senior High, CCIS, CCJE, CAS, CBM, CTEAS, and CTHM. The legend uses color-coded labels for both evaluation and attendance rates. Key performance indicators include: Total Events (5), Total Attendees (335), Attendance Rate (87%), Average Evaluation (89%), Active Users (400), and Total Users (5,000).

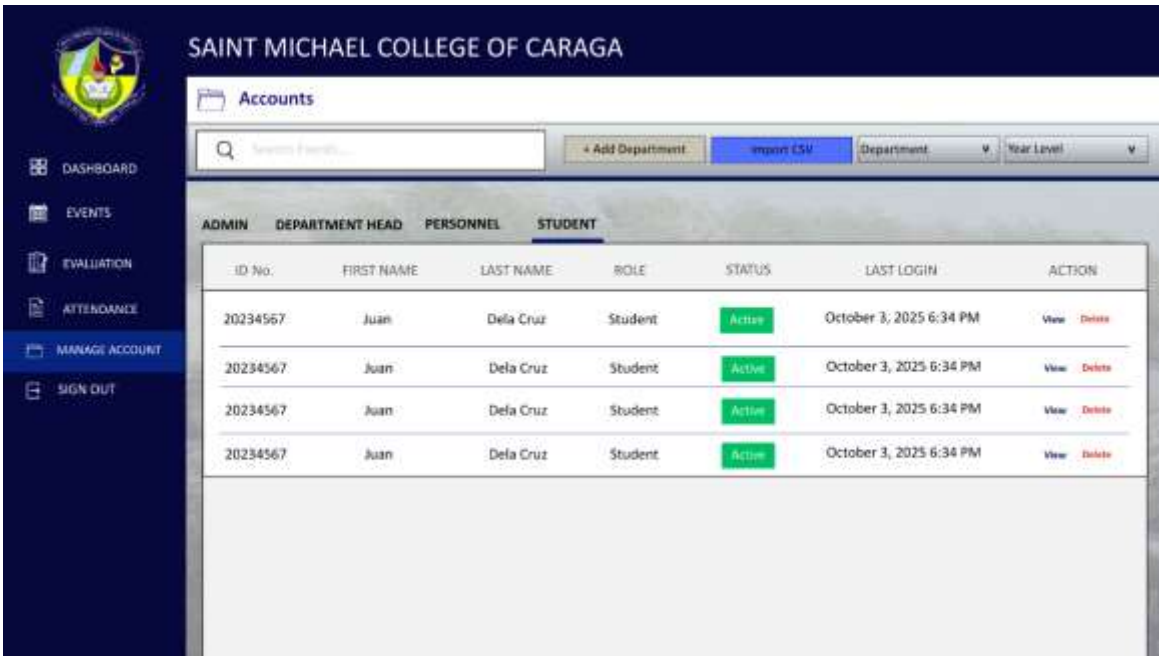


Figure 57. Student Account Management Interface-Admin

Figure 57 shows a student account management interface listing identical entries for a user named Juan Dela Cruz, with ID No. 20234567. Each row includes the student's role (Student), status (Active, highlighted in green), last login timestamp (October 3, 2025, 6:34 PM), and action links for "View" and "Delete." Above the table are controls for adding departments, importing CSV files, and filtering by department and year level.



Figure 58. Student Profile Display Interface-Admin

Figure 58 shows a student account profile interface under the “STUDENT” tab of the account management system. It displays individual student details including ID No. 202350946, First Name (Juan), Middle Name (Dela), Last Name (Cruz), Course (BSIT), and Year Level (3rd Year). The top section includes controls for searching events, adding departments, importing CSV files, and filtering by department and year level.



Figure 59. Department Account Creation Interface-Admin

Figure 59 displays a department-level account creation interface within the Saint Michael College of Caraga's management system. The form includes input fields for "Department Name" and "Password," accompanied by "Cancel" and "Create" buttons for user control. The background features a faded image of the college campus, reinforcing institutional branding.

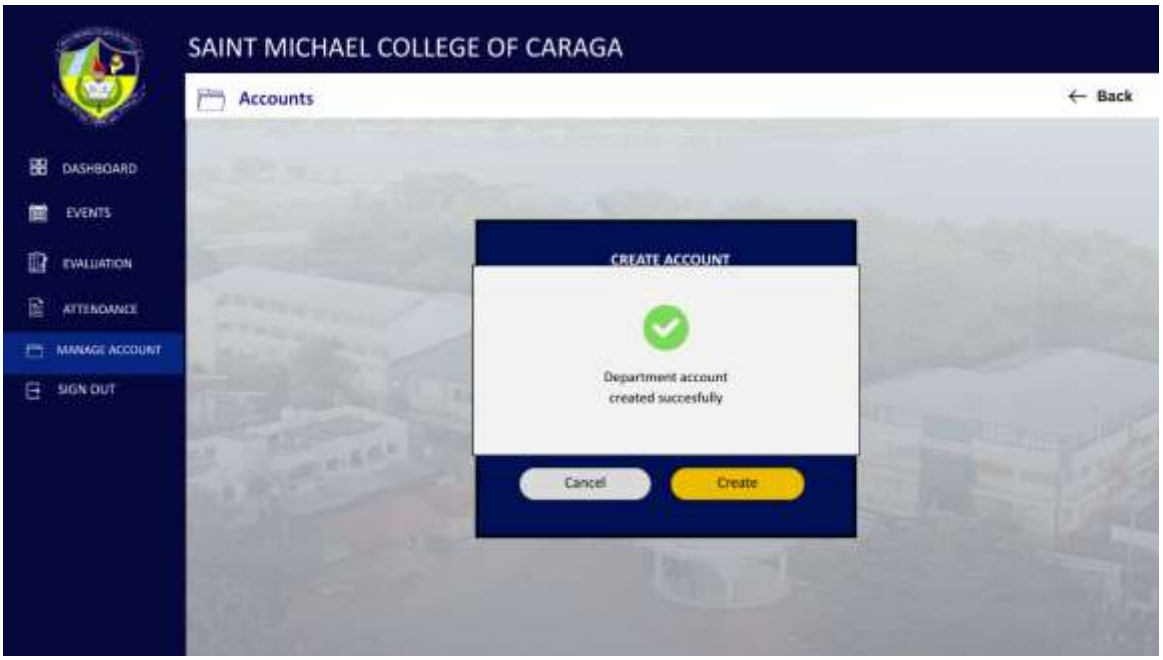


Figure 60. Department Account Creation Confirmation Interface-Admin

Figure 60 displays a confirmation dialog indicating successful creation of a department account within the Saint Michael College of Caraga's account management system. A green checkmark icon accompanies the message "Department account created successfully." Below the message are "Cancel" and "Create" buttons, with "Create" highlighted in yellow.



Figure 61. Departmental Evaluation and Attendance Report Interface-Admin

Figure 61 displays a statistical report interface titled “Evaluation Report (Per Event)” from Saint Michael College of Caraga. It includes filter options for Department and Date, and a button to download the report in PDF format. The “Overall Statistics” section displays key metrics: Total Events (5), Total Attendees (335), Attendance Rate (87%), Active Users (400), Total Users (5,000), and Average Evaluation (89%). The “Evaluation and Attendance per Department” table shows equal distribution across four departments—Elementary, Junior High, Senior High, and CCIS—each with 11.10% evaluation and attendance rates, one event, and attendee counts ranging from 45 to 60.

2.10 Software Platforms, Development Environment and Tools

Table 1

Software Platforms, Development Environment and Tools

Components	Specification	Usage
Internet Browser	Any	A web browser enables users to easily access and display content from the internet. It is responsible for loading the system's interface allowing users access to the system's features after a successful login.
Front-end	HTML	Defines the structural layout of the system's interface, arranging elements such as forms, text, and other key components.
	CSS	Improves the appearance and design of the HTML structure, providing a visually appealing and responsive interface.
	Javascript	Enhances interactivity by managing client-side functionalities, ensuring a dynamic and user-friendly experience.
	Bootstrap	Bootstrap allows developers to easily design responsive and mobile-first web pages by providing a collection of pre-designed components, grids, and styles to provide a more responsive website regardless of the screen size and device.
	jQuery	jQuery is a quick and efficient JavaScript library that simplifies HTML document manipulation, event handling, animations, and Ajax interactions. It enables developers to write less code for common web tasks and ensures cross-browser compatibility.

Back-end	MySQL (Database) PHP	MySQL powers data management by providing a robust database, allowing developers to store and manage information efficiently. By utilizing the interface of phpMyAdmin, developers can manage their databases and tables and can also perform queries. PHP is used on the server side which manages the interactions between user inputs and the database ensuring smooth data flow.
Server	Apache	Utilizing the XAMPP control panel, developers can manage the behavioral state of its components by ensuring smooth and easy access to local host servers like Apache and MySQL.
API	Axios	Axios is a JavaScript library used for handling API requests. It enables smooth communication between the client and server by sending and receiving data (e.g., JSON). In the ISO IQA Management System, Axios ensures reliable data transfer such as login authentication, retrieving records, and submitting evaluations, allowing users to interact with the system's backend services seamlessly.
QR Scanner	Built-in	Enables the system to automatically read and process QR codes using the device's camera or an integrated scanning module. It facilitates quick identification, attendance logging, or record retrieval without the need for external scanning applications.

2.11 Hardware Requirements

Table 2

Hardare Requirements

Components	Specification	Usage
Device	RAM (8 GB)	RAM facilitates fast and temporary data storage. It is used to enhance the speed and performance of devices in accessing information.
	Processor (Intel Core i5)	The processor or the CPU acts as the central component of the computer, handling the execution of instructions sent from the website. It also facilitates the processing of data.
	SSD (128 GB)	Devices employ SSD for storing large amounts of audit files, system software, and reports, while flash storage is commonly used in mobile devices for lightweight data storage. These storage solutions ensure quick retrieval and secure long-term data handling.
	Flash Storage (8 GB)	
Printer	Inkjet	The printer plays a vital role in both personal and professional settings, serving as a bridge between digital content and physical output. Here are the most common and practical uses of a printer.

2.12 System Evaluation

This section describes the different evaluation procedures done to the system to assess its functional suitability, performance efficiency, and usability. The survey instrument is adapted from ISO 25010 Software Product Quality Standards; the test scale gives four options to the respondents and will rate the system based on the question with 4 as the highest and 1 as the lowest.

Data were analyzed and interpreted based on the following parameters:

Table 3

Functional Suitability

Adjectival Rating	Scale Range (Mean)	Verbal Interpretation
4	3.50 – 4.0	Very Functional
3	2.50 – 3.49	Functional
2	1.50 – 2.49	Moderately Functional
1	1.0 – 1.49	Poor Functional

Table 3 presents the criteria used to evaluate a product or system's functional suitability according to the ISO survey.

Table 4

Performance Efficiency

Adjectival Rating	Scale Range (Mean)	Verbal Interpretation
4	3.50 – 4.0	Very Efficient
3	2.50 – 3.49	Efficient
2	1.50 – 2.49	Moderately Efficient
1	1.0 – 1.49	Poor Efficient

Table 4 outlines the scoring parameters used to assess the system's overall performance efficiency in executing its functions and tasks.

Table 5

Usability

Adjectival Rating	Scale Range (Mean)	Verbal Interpretation
4	3.50 – 4.0	Very Usable
3	2.50 – 3.49	Usable
2	1.50 – 2.49	Moderately Usable
1	1.0 – 1.49	Poor Usable

Table 5 presents the scoring parameters used to the usability of the system.

Table 6

Respondent's Distribution

SMCC		
Position	N (Population)	Percentage (%)
Personnel	1	12.5%
Student	7	87.5%
Total	8	100

Table 6 outlines the distribution of respondents who participated in evaluating the system within the SMCC. The respondents are categorized based on their positions, with 12.5% holding personnel roles and 87.5% being students. This indicates that most feedback comes from students directly interacting with the system. Since they are the primary users, their insights provide a more accurate assessment of the system's functionality, efficiency, and usability in real-world scenarios.

2.13 RESEARCH ETHICAL STANDARD

The researchers will make sure that the "Event Evaluation and Attendance System for Saint Michael College of Caraga" study is carried out with the utmost respect for participant rights, data privacy, and ethical principles. Throughout the research process, the following ethical considerations will be kept in mind.

A. Intellectual Property Rights (IPR)

The capstone project ensures the protection of intellectual property rights related to the Event Evaluation and Attendance System for Saint Michael College of Caraga (EEASFSMCC), developed for the Electronic Data Processing of Saint Michael College of Caraga. Since the EEASFSMCC involves original software code and system architecture, it may be protected under the Intellectual Property Code of the Philippines (RA 8293) through the Intellectual Property Office of the Philippines (IPOPhil), provided it meets the requirements of novelty, inventive step, and industrial applicability. The EEASFSMCC, including its design, instructional materials, and related documents, is protected under the Copyright provisions of the Intellectual Property Code of the Philippines (RA 8293). While the system's originality is implied, formal protection is secured through copyright registration with the Intellectual Property Office of the Philippines (IPOPhil). All third-party resources, templates, or software used in the development of the system are given proper citation and licensed accordingly**.** At the same time, the original components of the EEASFSMCC remain the intellectual property of the researcher under RA 8293 (Intellectual Property Code of the Philippines).

B. Informed Consent

The participants in this capstone project, who worked in the Electronic Data Processing department at Saint Michael College of Caraga, were fully informed about the purpose, extent, and potential outcomes of the EEASFSMCC study. Written consent was obtained from each

participant, ensuring their voluntary involvement and freedom to withdraw at any time without repercussions. The consent forms were presented in clear and accessible language, and participation occurred without any form of coercion. Personnel were informed about the data to be collected, including feedback on the EEASFSMCC and their use of technology, as well as how their responses would be used to improve the system

C. Data Privacy and Confidentiality

The capstone project strictly complies with the Data Privacy Act of 2012 (RA 10173) to safeguard the private and professional data of teachers. Data collected will be retained only for the duration necessary to fulfill the project's objectives, after which it will be securely disposed of. Furthermore, explicit consent is obtained from participants for any potential future use of their data. All information gathered, including survey responses and system usage, was anonymized through the removal of personal identifiers and the assignment of unique codes to protect participant identities. Digital data was stored in password-protected systems with encrypted storage, accessible only to authorized members of the research team under strict role-based permissions. Physical copies of the data, if any existed, were stored in locked cabinets. In the future, anonymized data may only be shared for academic or system development purposes with the participants' express consent.

D. Voluntary Participation and Freedom to Withdraw

All participants were informed that their involvement in the EEASFSMCC development and testing was completely voluntary. They were assured that they could withdraw at any time without repercussions, and that such withdrawal would not affect their professional roles or standing within the institution. To guarantee impartial and voluntary participation, no financial incentives, undue influence, or coercion were used.

E. Minimization of Harm and Risk Management

A risk analysis was conducted to ensure that participants would not experience any physical, psychological, professional, or academic harm. Risks were minimal since participation only involved the use and evaluation of the EEASFSMCC. In addition, risk management measures were implemented, including clear communication of procedures, monitoring of participant well-being, and the availability of support should any concerns arise. Responses and comments from participants were anonymized to prevent any negative professional effects. Participant support services were made available to address any issues that arose throughout the project. All activities were conducted with human participants only, and no animal testing was involved. The 3Rs principle (Replacement, Reduction, Refinement) did not apply because no animals were used in the study.

F. Beneficence and Contribution to Knowledge

This capstone project aims to provide the Electronic Data Processing personnel at Saint Michael College of Caraga with an effective Event Evaluation and Attendance System (EEASFSMCC) that streamlines event management, automates attendance tracking through QR code integration, and facilitates accurate evaluation and reporting. The system will enhance institutional operations by enabling role-based access, secure data handling, and efficient generation of attendance certificates, thereby supporting administrative functions and improving overall event documentation and monitoring. Results of the evaluation will be formally shared with institutional stakeholders to guide decision-making and future improvements. Participant sampling was conducted fairly, ensuring equal opportunity for involvement and unbiased representation across relevant groups. The system will enhance both academic and administrative processes by enabling role-based access, secure data handling, and efficient generation of attendance certificates. It will support institutional functions, improve event documentation and monitoring, and ensure that evaluation results are available to stakeholders upon request.

G. Justice and Fair Participant Selection

Personnel from Saint Michael College of Caraga who work in electronic data processing were selected because of their direct responsibility in managing technical operations and event-related data, making them the most relevant respondents for the study. Inclusion criteria required participants to hold roles connected to system deployment, data handling, or event management, and to voluntarily consent to involvement. Exclusion criteria applied to individuals without technical responsibilities or those unwilling to participate. The process ensured that no group was unfairly burdened, excluded, or overrepresented, maintaining transparency and equity throughout the project.

H. Data Integrity and Accuracy

All data related to the design, testing, and evaluation of the EEASFSMCC were collected with high accuracy and integrity. The research team followed standardized procedures to ensure that feedback and performance data were correctly documented and interpreted. Data reliability was further ensured through double-checking procedures, such as peer review of entries and the use of validation tools to cross-verify results. The research team followed standardized procedures to ensure that feedback and performance data were correctly documented and interpreted. To support the research process, ChatGPT was used during the preliminary analysis phase to organize thoughts and summarize initial insights Grammarly was used to ensure grammatical accuracy and clarity of the final research report. All AI-generated outputs were validated by the research team to prevent errors or misinterpretation.

I. Transparency and Honesty in Reporting

The evaluation and research findings of the EEASFSMCC will be presented openly and comprehensively. No data will be manipulated, and all relevant findings will be communicated with transparency. Both favorable and unfavorable results will be reported to ensure accuracy,

integrity, and accountability in the study. All third-party software and materials used in the system's development were properly cited to ensure transparency. Full acknowledgement of these sources was provided to maintain academic integrity and accountability.

J. Use of Patented or Copyrighted Materials

Although the EEASFSMCC created for this project is unique, All third-party software and materials used in the system's development were properly cited to ensure transparency. Open-source frameworks such as PHP, MySQL, and JavaScript were utilized in accordance with their respective licensing terms, and full acknowledgement of these sources was provided to maintain academic integrity, accountability, and compliance with open-source license requirements. Although no patented technologies were used in the development of the LMS, patent protection may be considered if it proves to be commercially viable. The project adhered to copyright compliance by ensuring no unauthorized materials were used in the system design or implementation.

K. Considerations for Human Subjects

Only human participants—students, staff, department heads, and administrators—were involved in this project. They were fully informed of their rights, the low risks, and the benefits of participating. Since no animals were used in the study, ethical clearance for animal involvement was not required. Any professional feedback provided was anonymized to protect the contributors' professional standing.

L. Responsible Use of AI and Other Technologies

This capstone project received ethical approval from the Institutional Review Board of Saint Michael College of Caraga on [insert date], confirming compliance with institutional standards for data privacy, intellectual property, and the ethical treatment of human participants. The system responsibly integrates technologies such as QR code scanning**,** role-based access**,** and

database encryption to ensure secure data handling. AI tools were used to assist in data synthesis, interface optimization, and automated reporting, with safeguards in place to prevent bias and ensure transparency. Plagiarism monitoring tools and validation protocols were employed to verify the originality and accuracy of all documentation and system-generated outputs.

M. **Ethical Clearance and Institutional Approval**

This capstone project received ethical approval from the Institutional Review Board of Saint Michael College of Caraga. Approval was granted on October 24, 2025 , ensuring compliance with all necessary guidelines regarding data privacy, intellectual property, and the treatment of human participants.

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